Toward an optimal data-generating process for simulating predictive-biomarker interactions in aggregated N-of-1 trials

pmsimstats team

November 07, 2026

Contents

Abstract **6**

Background **7**

Aggregated N-of-1 trials and predictive biomarkers . . . 8
The starting DGP . . . 9
Why the joint distribution is built from the design . . . 10
What a data-generating process for these trials must satisfy . . . 12
Aims . . . 13

Methods **13**

The starting data-generating process . . . 14
Estimands . . . 15
The paired-difference statistic . . . 16
The feasibility bound . . . 17
Simulation studies . . . 18

Results **22**

Stage 1. Feasibility of the joint distribution . . . 22
Stage 2. The biomarker-response correlation under carryover . . . 23
Stage 3. Carryover on the mean . . . 32
Stage 4. Serial correlation . . . 33
Stage 5. Cross-component structure . . . 35
Stage 6. Features the recommended DGP adds . . . 37
Stage 7. The recommended DGP . . . 42
Using the recommended DGP to plan a trial . . . 45

Discussion **52**
The correlation encoding of the interaction . . . 52
What remains uncalibrated . . . 54
Recommendations . . . 55
Limitations . . . 55

Conclusions **57**

Glossary **57**

Abbreviations **61**

Supplementary material **61**

Declarations **62**

References **63**

Appendix A. Closed-form derivations **71**
A.1 Moments of the paired-difference statistic . . . 71
A.2 Power . . . 71
A.3 Monotonicity under the decaying correlation . . . 72
A.4 The feasibility bound and the shared biomarker effect . . . 72
A.5 The separable structure . . . 74
A.6 The responder-class reading of c_{bm} . . . 76

Appendix B. Code changes **77**
B.1 The decaying correlation in the starting `generateData()` (commit 58b32a9) . . . 77
B.2 Stop instead of repairing . . . 78
B.3 Carryover decaying from the last on-drug mean . . . 79
B.4 The exposure-decayed drug indicator in the analysis . . . 81
B.5 The recommended DGP in pmsimstats-ng . . . 81

Appendix C. First run of the stress test **83**

List of Figures

1 Closed-form power (lines) against carryover half-life, with Monte Carlo estimates (points; 5,000 replicates per cell), under the constant rule (constant correlation after first exposure) and the decaying rule (correlation decaying at the carryover half-life). Hybrid design, $N = 70$, $c_{bm} = 0.25$, compound symmetry; left panel E9b, right panel E9. 24

2 Closed-form power against the on/off correlation difference, with the positions of the two correlation rules. Hybrid design, $N = 70$, $c_{bm} = 0.25$, E9. 26

3 Power to detect the biomarker-by-treatment interaction, in the layout of Figure 4B of the earlier study, from the starting code at commit 58b32a9 as published (upper two panel rows) and with the decaying correlation (lower two panel rows). Rows within panels: censoring patterns; columns: carryover half-life; panels: trial design (N-of-1 is the Hybrid design) and total sample size. $c_{bm} = 0.25$, 250 replicates per cell (Monte Carlo SE at most 0.032), the same seeds for both versions. No matrix was repaired in either version. 28

4 Power of the analysis G10 to detect the biomarker-by-treatment interaction under the recommended DGP (separable AR(1), $\rho = 0.7$, $c_1 = 0.2$, decaying correlation, carryover decaying from the last on-drug mean, original response parameters), in the layout of Figure 4B of the original study. Rows within panels: censoring patterns; columns: carryover half-life; panels: trial design (N-of-1 is the Hybrid design) and total sample size. $c_{bm} = 0.25$, 250 replicates per cell (Monte Carlo SE at most 0.031). Censoring patterns use the corrected dropout rule, under which dropout rises with worsening; the original-code figure uses the published rule of commit 58b32a9. Gray cells (n/e): the open-label design, in which every visit is on drug and the interaction is not estimable under G10. 30

5 Closed-form power of the paired-difference statistic by interaction strength (c_{bm}) and carryover half-life, in the layout of Figure 4B of the earlier study: the constant rule and the decaying correlation. Upper row: E9b, which counts baseline as an off-drug visit as the binary-indicator analysis does; lower row: E9. Gray cells: c_{bm} at or above the compound-symmetry feasibility bound, where the correlation matrix is not positive definite. Hybrid design, $N = 70$, compound symmetry. 31

6 Drug-response mean by visit in the Hybrid design, by path, under the anchored carryover rule (decay from the last on-drug mean, the recommended DGP) and the compounding carryover rule (compounded decay), at carryover half-lives of 0.2 and 1 week. Filled points: on-drug visits; open points: off-drug visits. D: drug; P: placebo. 33

List of Tables

1 The analyses used in this paper and the quantity each estimates. 16

2 Design of the three simulation studies in the ADEMP structure. MCSE: Monte Carlo standard error of a rejection rate, $\sqrt{p(1-p)/R}$ for R replicates. 18

3 Closed-form power by carryover half-life (weeks). Hybrid design, $N = 70$, $c_{bm} = 0.25$, compound symmetry. Constant: the constant rule (constant correlation after first exposure). Decaying: the decaying rule (correlation decaying at the carryover half-life). E9: paired-difference statistic on post-baseline visits; E9b: the same with baseline counted as an off-drug visit, as in the binary-indicator analysis. Against Monte Carlo estimates (5,000 replicates per cell), 2 of the 24 cell-variants in this table differ by more than 2 Monte Carlo SE, and 16 of all 566 cell-variants of study 1 (about 26 expected by chance). 24

4 The starting code at commit 58b32a9, as published (constant) and with the decaying correlation (decaying): Hybrid design, $N = 70$, $c_{bm} = 0.25$, 500 replicates per cell, same seeds. Monte Carlo SE of power at most 0.022. A rerun of the original version at 0.1 week with 4,000 new replicates gave power 0.090 (Stage 2 text). 26

5 Serial correlation: the feasibility bound on c_{bm} (Ceiling; decaying correlation, no carryover, minimum over the Hybrid, crossover and OL+BDC designs) and closed-form E9 power (Hybrid, $N = 70$, no carryover) by ρ, under compound symmetry and under the separable AR(1) form of Stage 5. Under compound symmetry the Hybrid and crossover designs tie for the minimum. Under the separable form the minimum is attained in the Hybrid design at $\rho = 0.8$ and in the OL+BDC design at $\rho \leq 0.7$, and the crossover design is never binding. At $t_{1/2} = 0.2$ week every power value is lower by 0.004 to 0.009. 34

6 Serial correlation under the separable AR(1) form of the recommended DGP ($c_1 = 0.2$, decaying correlation, carryover decaying from the last on-drug mean): the feasibility bound on c_{bm}, minimum over the Hybrid, crossover and OL+BDC designs, by carryover half-life, the design that attains the minimum without carryover, and the closed-form total N for 80 percent power of E9 in the Hybrid design at the 0.2-week half-life. Not feasible: c_{bm} lies above the Hybrid bound. 35

7 Largest feasible biomarker-response correlation c_{bm} under the decaying correlation, without carryover, minimum over the Hybrid, crossover and OL+BDC designs. Under compound symmetry the Hybrid and crossover designs tie for the minimum. Under the two AR(1) forms the minimum is attained in the Hybrid design at $\rho = 0.8$ and in the OL+BDC design at $\rho = 0.7$ (separable form: Hybrid 0.480), and the crossover design is never binding. 37

8 Null rejection rates under a prognostic biomarker ($c_{bm} = 0$). Each pathway has strength 0.3: a correlation of 0.3 with the natural-history component (constant), the placebo-response component or baseline severity, and for the growing natural-history pathway a mean shift of the natural-history component of 0.3 of its SD (3 points) per SD of biomarker at the end of its curve, growing along the curve. 1,000 replicates per cell (two-MCSE band 0.036 to 0.064 for a single cell). Third run of study 3: the recommended DGP, with carryover decaying from the last on-drug mean and fixed path allocation. Rates above 0.075 in bold. The first run is in Table 17; the changes between runs are given in the text. 38

9 Largest feasible prognostic correlation given the biomarker-response correlation, and the reverse, under the recommended DGP (separable AR(1), $\rho = 0.7$, $c_1 = 0.2$), Hybrid design, minimum over its four paths. Correlation: the quantity whose bound is given. c_{PB}, c_{TV}, c_{BL}: correlation of the biomarker with the placebo-response component, the natural-history component and baseline severity. None: no value is feasible. 40

10 The optional features of the recommended DGP: null rejection rate ($c_{bm} = 0$; 1,000 replicates, MCSE 0.007) and power ($c_{bm} = 0.30$; 500 replicates, MCSE at most 0.022) for the binary-indicator analysis (Binary), the random-intercept CAR(1) model with CR2 and the expectancy-term model with CR2 (Expect.). Hybrid design, third run of study 3 (the recommended DGP: carryover decaying from the last on-drug mean, fixed allocation). Each row changes one feature of the base cell. 42

11 The recommended data-generating process. The Default column gives the value of each element in the recommended DGP; where the stress test of Stage 6 used another value it is stated. The stress test's reported run uses fixed allocation; its second run assigned participants to paths at random, a driver option outside the recommended DGP (Appendix B.5). 43

12 Planning under the recommended DGP (separable AR(1), $\rho = 0.7$, $c_1 = 0.2$, decaying correlation, carryover decaying from the last on-drug mean, starting response parameters): total N for 80 percent power of the paired-difference statistic E9 at $\alpha = 0.05$, by design, interaction strength and carryover half-life, with the implied slope in symptom points per SD of biomarker ($\sigma_{BR} = 8$) and the power at $N = 70$ and the 0.2-week half-life for E9 (closed form) and for the analysis G10 (simulated, 1,000 replicates per cell, Monte Carlo SE at most 0.016). Participants are allocated to paths in the original fixed proportions. 49

13 Rejection rate of the analysis G10 under the recommended DGP of the preceding planning table, by design, interaction strength and carryover half-life, at $N = 70$ and at the total N the closed-form E9 rule gives for 80 percent power (the preceding table). Rows with $c_{bm} = 0$ give the size at $N = 70$ (2,000 replicates per cell, Monte Carlo SE at most 0.005); the other rows give power (1,000 replicates per cell, Monte Carlo SE at most 0.016). †: the E9 sample size exceeds 400 and the cell ran at $N = 400$. Rejection rates are over converged fits (at most 0.8 percent of replicates failed in any cell). 49

14 Total N for 80 percent power in the Hybrid design under the recommended DGP with the starting parameters: closed-form E9 (Table 12; at $\rho = 0.5$ and 0.9, Table 15) and the analysis G10. G10 was simulated at 6 values of N per cell at $\rho = 0.7$ (500 replicates each), at the same multiples of the E9 value at 0.5, and at lower multiples at 0.9, where G10 needs far fewer participants than E9. The N at which a probit regression of rejection on $\log N$ reaches 0.80 is rounded up. 50

15 Planning by serial correlation at the 0.2-week half-life under the recommended DGP (separable AR(1), $c_1 = 0.2$, starting response parameters): the closed-form total N for 80 percent power of E9, and G10 power at $N = 70$ and at that N, with the percent of G10 fits that failed at $N = 70$. 1,000 replicates per cell. The rows at $\rho = 0.7$ repeat Table 12. At $\rho = 0.9$ the grid of c_{bm} lies inside the lower bound. Not feasible: c_{bm} lies above the bound of that design. †: the E9 sample size exceeds 400 and the cell ran at $N = 400$. Power is over the fits that succeeded. 51

16 Correlation matrices that are not positive definite. Figure 4 grid: open label, open label plus blinded discontinuation, crossover and Hybrid designs; $t_{1/2} \in \{0, 0.1, 0.2\}$ week. Extended grid: the last three designs; $t_{1/2} \in \{0, 0.1, 0.2, 0.5, 1\}$ week. The starting-DGP counts reproduce those of the starting code exactly. The recommended-DGP counts are the same at $\rho = 0.7$ (the recommended value) and $\rho = 0.8$ because both feasibility bounds lie between 0.3 and 0.6. 80

17 Null rejection rates under a prognostic biomarker in the first run of study 3 (compounding carryover rule, fixed allocation), 1,000 replicates per cell (two-MCSE band 0.036 to 0.064 for a single cell). Rates above 0.075 in bold. 83

Abstract

Background. A predictive biomarker is validated by an interaction test. Unfortunately, such tests are expensive. With two equal subgroups and an interaction as large as the main effect, the test needs about four times the sample of a main-effect test (Brookes et al. 2004). Aggregated N-of-1 designs have been proposed as a way to afford it. Their power must be estimated by simulation. A simulation can only show what its data-generating process (DGP) allows. But what should that process be? We set out five criteria for a DGP for these trials. We then take its elements one at a time and compare the candidate choices. We start from the one DGP we found that encodes a predictive biomarker (Hendrickson et al. 2020). It draws the biomarker and three latent response components from one multivariate normal distribution. The biomarker enters as a correlation with the drug response.

Methods. We derive the closed-form moments and power of a paired-difference interaction statistic. We also derive a Schur-complement bound on the largest feasible interaction. We check these results by Monte Carlo simulation. Lastly, we fit fourteen analyses against a prognostic biomarker and report ten of them. The Hybrid design is the primary case.

Results. The first element is the correlation at off-drug visits. If it stays constant after first exposure, on-drug and off-drug visits carry the same moderation under any carryover. Power then falls nearly to the test size. If it decays at the carryover half-life, power returns to its no-carryover

value at half-lives of 0.1 to 0.2 week. It declines at longer half-lives. The decline is proved monotone for the paired-difference statistic within each design path. In simulations of the starting code at 0.1 week, power was 0.09 with the constant correlation (4,000 replicates). It was 0.64 with the decaying one, against 0.67 without carryover. The second element is the serial correlation structure. It sets the largest interaction the DGP can hold. Under compound symmetry no joint distribution exists for correlations above 0.256. A separable AR(1) structure raises the bound to 0.474 at a serial correlation of 0.7, a conventional value. A PTSD symptom series suggests a slower decay, however. At a serial correlation of 0.9 the bound falls to 0.34, and the Hybrid design needs about half the sample it needs at 0.7. Finally, a prognostic biomarker may act through a pathway absent at baseline. Analyses that share the biomarker effect with baseline are then anticonservative.

Conclusions. We recommend a DGP with three features: a biomarker-response correlation that decays after discontinuation, a separable AR(1) structure, and a check that refuses infeasible effect sizes. We give closed-form results, code and planning sample sizes under it. Its parameter values are not yet calibrated. The recommendation is therefore a step toward an optimal DGP, not the end point.

Trial registration. Not applicable.

Keywords: N-of-1 trials; predictive biomarker; simulation; carryover; data-generating process; power; prazosin.

Background

Validating a predictive biomarker requires a biomarker-by-treatment interaction test (1). Unfortunately, such a test is expensive. When the interaction is as large as the main effect and divides the sample into two equal subgroups, it needs about four times the sample of the main-effect test (2). Aggregated N-of-1 designs have been proposed as a way to afford it. Their power must be estimated by simulation.

A simulation can show only what its data-generating process (DGP) allows. The DGP decides how the biomarker acts, how the drug effect builds and wanes, and how one period carries into the next. Yet there is little guidance on what a DGP for these trials should be. In this paper we explore that question. We set out five criteria a DGP should meet. We then take its elements one at a time. For each element we compare candidate choices against the criteria and keep the better one. The

result is a recommended DGP. We do not claim it is optimal. We claim it is closer to optimal than its starting point, and we say which of its parts remain uncalibrated.

The starting point is the DGP of Hendrickson et al. (3), which we call the starting DGP. It is the only DGP we found that encodes a predictive biomarker for these designs. It draws the baseline biomarker and three latent response components at every visit from one multivariate normal distribution. The trial design assembles the correlation matrix from each participant's exposure state. The biomarker enters as a correlation with the drug response. The primary case is the Hybrid design. It combines open-label titration, blinded discontinuation and a brief crossover, with carryover between phases.

Aggregated N-of-1 trials and predictive biomarkers

In an aggregated N-of-1 trial each participant moves between drug and placebo, or between drug and no drug, several times. The individual series are analyzed together (4–6). Every participant contributes both an on-drug and an off-drug response. The design can therefore estimate how the drug effect varies between people. It can then test whether a baseline measurement predicts who benefits (7,8). The motivating example is prazosin for post-traumatic stress disorder (PTSD). Prazosin has been tested in several randomized trials (9–11). The largest, the VA Cooperative Studies Program trial CSP 563 (304 veterans), found no benefit on its primary outcomes (11). In a secondary analysis of an earlier trial in 67 soldiers, higher pretreatment standing systolic blood pressure was associated with greater symptom reduction on prazosin but not on placebo, about 14 points on the Clinician-Administered PTSD Scale per 10 mm Hg (12). Blood pressure was therefore proposed as a candidate predictive biomarker. That association did not replicate in our reanalysis of CSP 563 (Additional file 1; participant-level data not public). In 279 participants the arm-by-blood-pressure interaction was +0.25 points per 10 mm Hg (95% CI −2.72 to 3.21), which excludes the published value. Prazosin and blood pressure therefore serve here as a worked example for the units of the planning section, not as an established predictive biomarker.

Recent reviews cover the design and analysis of N-of-1 trials, including Bayesian models for series of trials (13,14). Closed-form power calculations exist only for simplified versions of these designs (15,16). Power is therefore estimated by simulating trials from a data-generating process (DGP) and analyzing each simulated trial (17). Several simulators of N-of-1 and aggregated N-of-1 data exist. Percha et al. (18) simulate single series with exponential wash-in and wash-out to choose block lengths and sampling rates. Chen and Chen (19) generate cycles under compound symmetry

or AR(1) with a carryover effect to compare analyses. Blackston et al. (20) and Gärtner et al. (21) simulate aggregated series with carryover to compare designs and estimators. None of these four represents a baseline biomarker that modifies the drug response. Tueller et al. (22) estimate power for idiographic within-subject trials by simulation. Grenet et al. (23) simulate crossover and parallel-group trials with a binary predictive biomarker. They encode it as a mean-term interaction. In single-case research, multilevel models with moderators of the intervention effect are established (24,25), and Moeyaert et al. (26) estimate by simulation the power to detect such a moderator. There, too, the moderator enters the mean model.

The starting DGP

The starting DGP has three parts. The first is a component decomposition of the symptom score. The score is the baseline level minus three responses. Each response follows a modified Gompertz curve. The drug-response component (BR) is driven by time since the current on-drug period began. The placebo-response component (PB) is driven by time in the trial and scaled by the participant's expectancy, the probability of receiving drug given the design. The natural-history component (TV) is driven by time in the trial. The second part is the design specification for open-label, crossover, blinded-discontinuation and Hybrid designs. It tracks time on drug, time since discontinuation and expectancy at each visit. The third part is open software (the `pmsimstats` R package).

The starting DGP also encodes the predictive biomarker in a distinctive way. Statistical models and sample-size methods for N-of-1 and crossover designs usually represent treatment-effect heterogeneity as random participant-by-treatment variation (7,27–31). The starting DGP instead places the biomarker in the same multivariate normal distribution as the response components. The biomarker has a nonzero correlation c_{bm} with the drug response at on-drug visits. Without carryover it has zero correlation elsewhere. We call c_{bm} the biomarker-response correlation. A predictive biomarker, a moderator of the treatment effect (32), is then one whose association with the outcome differs between on-drug and off-drug visits. The parameterization keeps the marginal distribution of the drug response fixed. It expresses the strength of the interaction as a correlation. We call this the correlation encoding of the interaction.

The statement that the encoding has no precedent rests on three database searches. On 2026-10-27 we searched PubMed for (N-of-1 OR single patient OR aggregated N-of-1) AND simulation AND (biomarker OR effect modifier OR treatment-by-covariate) AND (power OR sample size). It

returned one record, the study that proposed the starting DGP. On 2026-11-03 we searched PubMed titles and abstracts for ("N-of-1" OR "single-patient" OR "single-case" OR "aggregated N-of-1" OR crossover) AND (simulation OR simulated OR "Monte Carlo") AND (biomarker OR moderator OR "effect modifier" OR "treatment-by-covariate") AND (power OR "sample size"). We ran the same terms, without "Monte Carlo", against abstracts in Europe PMC. Both returned two records: that study and a simulation of moderator power in single-case multilevel models (26). Neither search retrieved Grenet et al. (23), which we identified otherwise, so the searches are not exhaustive. Web of Science and Scopus were not searched. The searches cover the N-of-1, crossover and single-case literature indexed in these two databases. They do not cover the wider literature on joint simulation of covariates and outcomes, such as copula methods. In that literature, generating variables jointly from a specified correlation matrix, and checking that the matrix is feasible, is standard (33,34). What we found no precedent for is a biomarker-by-treatment interaction encoded as a correlation whose value depends on the design and the exposure state. The claim is therefore limited to the N-of-1 literature we searched.

Why the joint distribution is built from the design

The simplest DGP that could serve would generate one symptom score per visit. Its mean would carry the design effects and an interaction term $\beta\, B_i D_{it}$. Its residual would be compound symmetric or AR(1). Most simulators of crossover and N-of-1 data use that form (15,19), and the analysis model assumes it. It is adequate when the question is the main effect. For the questions this series asks, the joint-distribution form of the starting DGP differs from it in four ways. The simple form stays usable: a mean model with a random participant-by-treatment slope (7,28) is the conditional equivalent of the joint-distribution form (Discussion).

1. **How the moderation persists after discontinuation is stated separately from the mean carryover.** Carryover in crossover trials is usually modeled as a mean effect (35), a model whose usefulness has been questioned (36,37). In the simple form the analyst adds it as a term such as $\beta B_i D_{bc,it}$. In the joint-distribution form it is the correlation profile g_t. Either is a modeling choice. The joint-distribution form makes the choice visible in the same quantity that sets the effect size.
2. **The interaction is expressed as a share of between-person variation.** With the marginal variance of the drug response held fixed, c_{bm}^2 is the share of the on-drug drug-response

variance that the biomarker explains linearly. Appendix A.6 relates it to a responder-class model. A mean model with a random participant-by-treatment slope implies the same kind of quantity through the ratio of the fixed slope to the slope variance. Neither form makes the residual variance depend on the biomarker.

3. **Prognostic pathways are correlations in the same matrix.** A biomarker may also be associated with the placebo response or with the natural history of the condition. In the joint-distribution form such a pathway is a further correlation in the biomarker row, and it shares one feasibility condition with c_{bm} (Stage 6). A pathway can also be generated as a mean shift, as the stress test does for the growing natural-history pathway (Appendix B.5).
4. **Existence becomes a question with an answer.** In the simple form every parameter combination yields a valid distribution. In the joint-distribution form, with the marginal variances fixed, it does not. Whether the simulated effect sizes are jointly possible, and how much of the interaction the design can see, then have closed-form answers.

The starting DGP specifies it. The biomarker and the three response components at every visit form one multivariate normal vector. The design assembles its correlation matrix. The biomarker is correlated with the drug response at on-drug visits, and at a value set by the exposure state at off-drug visits. The components are serially correlated in calendar weeks. The components are cross-correlated through a common block. Three results follow. The largest feasible interaction has a closed form. It depends on the design and on two modeling choices: the marginal drug-response variance is held fixed, and the drug-response component keeps its full variance and serial correlation at off-drug and pre-exposure visits. We call it the feasibility bound. The power of a paired-difference statistic has a closed form. The design enters it through the on/off correlation difference, the on-drug correlation minus the mean off-drug correlation. The residual structure the analysis model should assume can be derived from the DGP. Three requirements come with this form. The marginal covariance of the outcome is implied rather than chosen. The joint distribution must exist. Every entry must be calibrated or declared uncalibrated. The Discussion shows that, conditional on the biomarker, the correlation encoding reduces to the mean-term form with a smaller residual variance, so the standard form remains available within it. We keep the correlation encoding because it is the starting point. We do not claim it is preferable to a mean-term or random-coefficient form.

The DGP is not new in its parts. Simulating repeated measures from a multivariate normal with compound symmetry or AR(1) and a carryover term is routine (15,19). Decomposing a clinical tra-

jectory into disease progression, placebo response and drug action is the pharmacometric convention (38). Separable (Kronecker) covariances for several response variables measured repeatedly are an established model (39,40). The starting DGP combined these parts. It placed a baseline variable in the joint distribution of the latent response components, with a correlation indexed by each visit's exposure state. Designs for heterogeneity analyses in cluster randomized trials also use correlations of an effect modifier, but these are intraclass correlations of the modifier and of the outcome, and the modification itself is a term in the mean model (41). This paper refines that combination. We found no other instance of it in the N-of-1 literature we searched (above).

What a data-generating process for these trials must satisfy

To compare candidate choices we need a criterion for "better". We use five. Each stage of the Results takes one element of the DGP. It compares the candidates and ends with the choice and the criterion it rests on.

1. **Existence.** The joint distribution must exist for the effect sizes a study simulates. The correlation matrix must be positive definite. We measure this by the largest feasible c_{bm}, the feasibility bound.
2. **Identifiability under carryover.** The biomarker-by-treatment interaction must remain a contrast between on-drug and off-drug visits when carryover is present. We measure this by the on/off correlation difference. The closed form shows that this difference governs power.
3. **Fidelity to the pharmacology.** The time course of the drug effect, its decay after discontinuation and its behavior at re-exposure should be consistent with a short-half-life, titrated drug such as prazosin. This is a judgment. We state it as such where it decides a stage.
4. **Analyzability.** The residual structure the DGP generates should be one the analysis model can represent. The DGP should also be able to generate the features that decide which analyses are valid, such as a prognostic biomarker. We measure this by the agreement of the generated residual structure with the analysis model's, and by null rejection rates under those features.
5. **Calibration status.** Every parameter should be labeled as the starting value, a choice made here, or uncalibrated. The DGP's values have not been re-estimated for this paper. A reanalysis of CSP 563 (planning section) estimates some quantities on the total symptom

score and shows that others are not identifiable from a parallel-group trial. This criterion is therefore reported, not met.

A DGP meeting criteria 1 to 4, with criterion 5 reported, is what we call the recommended DGP. Its label in the repository's code and data is `refined`.

The reference drug. Prazosin's plasma half-life is about 2.3 hours (42), against weekly visits. The plasma concentration therefore falls by orders of magnitude within a day of the last dose. We assume a symptom-level half-life of 0.1 to 0.2 week (17 to 34 hours), or 7 to 15 plasma half-lives, as the pharmacological range. No symptom-response half-life for prazosin in PTSD has been reported, so the range is an assumption. A drug's effect need not decline with its plasma concentration (43), so the plasma half-life is only a starting point. The Results therefore cover 0 to 2 weeks and label half-lives of 0.5 week and above as behavioral scenarios. The drug is re-titrated after an interruption, which bears on how the DGP treats re-exposure (Stage 3).

Aims

We have three aims. First, we judge the elements of a DGP for these trials against the five criteria, one element at a time. The elements are the handling of infeasible matrices, the biomarker-response correlation off drug, carryover on the mean, the serial correlation, the cross-component structure, and the optional features that decide which analyses are valid. Second, we derive in closed form what each candidate does to feasibility and to power. Third, we collect the choices as a recommended DGP and give the sample sizes a trialist needs to plan under it. The Hybrid design is the primary case. The open-label, blinded-discontinuation and crossover designs are comparators.

Methods

The Methods describe the starting DGP and the tools used to compare candidate choices: the estimands, a statistic with a closed-form distribution, the feasibility bound and the simulation studies. Every choice about the DGP is made in the Results.

The starting data-generating process

We use the starting DGP's notation, with a few additions. Participant i is observed at a baseline visit $t = 0$ and at post-baseline visits $t = 1, \ldots, T$, at calendar weeks w_t (4, 8, 9, 10, 11, 12, 16 and 20 in the Hybrid design). The symptom score is

$$Y_{it} = \mathrm{BL}_i - (TV_{it} + PB_{it} + BR_{it}),$$

so a larger response lowers the score. Each component has a mean given by the modified Gompertz curve

$$G(x;\, m, d, r) = m\,\frac{\exp(-de^{-rx}) - e^{-d}}{1 - e^{-d}},$$

which the code obtains by subtracting the offset me^{-d} from $m\exp(-de^{-rx})$ and rescaling, so that $G(0) = 0$ and $G(\infty) = m$. It is evaluated at a component-specific clock:

- The drug-response component (BR) at time on drug t_{od}, the time since the current on-drug period began, with $m = 10.99$, $d = 5$, $r = 0.42$ and SD 8. The curve reaches its inflection at 3.8 weeks, half its maximum at 4.7 weeks and 90% at 9.2 weeks.
- The placebo-response component (PB) at time in the trial, scaled by the expectancy η_t (1 in the open-label phase, 0.5 when blinded), with $m = 6.51$, $d = 5$, $r = 0.35$ and SD $10\eta_t$.
- The natural-history component (TV) at time in the trial, with the same curve as PB and SD 10.

The biomarker has mean 124.3 and SD 15.4, and baseline severity has mean 83.1 and SD 18.5. These are the values of the starting code's parameter objects at commit 58b32a9, stored in the package as `extracted_rp` and `extracted_bp`. Every simulation and closed-form calculation in this paper uses them, including the simulations of the analysis G10 in the planning section and the paired rerun of Appendix B.4. In those two, the biomarker enters the analysis standardized to mean 0 and SD 1, so their interaction coefficients are per SD of biomarker. The standardization is linear and does not change the test.

Off drug, the drug-response mean decays from its previous value with half-life $t_{1/2}$: $\mu_{BR,t} = \mu_{BR,t-1}\, 2^{-t_{sd,t}/t_{1/2}}$, where t_{sd} is the time since discontinuation. We call this the compounding

carryover rule. When the drug is resumed, time on drug restarts at zero.

The vector (biomarker, baseline, $TV_{i\cdot}$, $PB_{i\cdot}$, $BR_{i\cdot}$) is multivariate normal. Within each component the correlation between any two visits is $\rho_{cs} = 0.8$ (compound symmetry). We write ρ_{cs} to keep ρ for the AR(1) correlation per week. Between components the correlation is $c_1 = 0.2$ at the same visit and $c_\times = 0.1$ otherwise. The biomarker's correlation with BR_{it} is $c_{bm}\, g_t$, where g_t is the correlation profile, the fraction of c_{bm} present at visit t. The constant rule sets $g_t = 1$ wherever the drug-response mean is nonzero and 0 elsewhere. The correlation is therefore constant after first exposure. When the correlation matrix is not positive definite, the starting code replaces it with a nearby valid matrix, without a warning (Appendix B.2).

The binary-indicator analysis is the linear mixed model

$$Y_{it} = \beta_0 + \beta_B B_i + \beta_D D_{it} + \beta_T w_t + \beta_{bm:D} B_i D_{it} + u_i + \varepsilon_{it},$$

with a random intercept u_i, the binary drug indicator D_{it}, and baseline included as a row with $D_{i0} = 0$. The interaction $\beta_{bm:D}$ is tested with Satterthwaite degrees of freedom (44). Participants are assigned to the design's paths in fixed numbers (18, 18, 17 and 17 in the Hybrid design at $N = 70$), not at random.

Estimands

The target is the on-drug biomarker moderation of the drug response. This is the change in the drug effect per unit of biomarker while the drug is fully present. Without carryover it equals $\beta_{bm:D} = -c_{bm}\sigma_{BR}/\sigma_{bm}$.

Analyses that contrast on-drug with off-drug visits through a binary drug indicator estimate this quantity only when no moderation remains off drug. Under carryover they estimate an attenuated on-minus-off contrast. Analyses with an exposure-decayed indicator $D_{bc,it}$ at the true half-life recover the on-drug moderation, because the decaying correlation and $D_{bc,it}$ share one exponential decay. Under the Weibull decay option, or with a correlation half-life that differs from the carryover half-life, they do not. At another analysis half-life, fixed or chosen from the data, they recover it only approximately. Table 1 states which analysis in this paper targets which quantity. The three sets of results are compared only where their targets coincide (no carryover) or where the attenuation is the object of study (Stage 2).

Table 1: The analyses used in this paper and the quantity each estimates.

Analysis	Drug indicator	Estimates	Used in
Paired-difference statistic E9, E9b (closed form)	binary	on-minus-off contrast, $\beta_{bm:D}$ times the on/off correlation difference	Stages 1, 2, 4, 5; planning
The binary-indicator analysis (mixed model)	binary	the same contrast	Stage 2 (starting code)
Stress test: the binary-indicator analysis	binary	the on-minus-off contrast	Stage 6
Stress test: fixed-half-life analyses	$D_{bc,it}$ at an analysis half-life of 0.5 week, the true half-life except in the cells that vary it	on-drug moderation $\beta_{bm:D}$ where the half-lives match	Stage 6
Stress test: analyses with the half-life chosen by AIC	$D_{bc,it}$ at the AIC-chosen half-life (0, 0.25, 0.5, 1 or 2 weeks)	on-drug moderation $\beta_{bm:D}$ where the chosen half-life is the true one	Stage 6

The paired-difference statistic

To isolate how carryover acts, we use a summary statistic whose distribution can be derived exactly (Appendix A). For each participant,

$$\Delta_i = \bar{Y}_i^{\text{on}} - \bar{Y}_i^{\text{off}} = \sum_t a_t Y_{it},$$

the difference between the mean on-drug and mean off-drug scores. It is regressed on B_i by least squares. We call it E9 when it uses post-baseline visits only. We call it E9b when baseline is counted

as one more off-drug visit, as in the binary-indicator analysis. In the Hybrid design, Monte Carlo simulation of the binary-indicator mixed model shows that E9b reproduces its mean estimates to within 0.004. It reproduces its power closely (0.654 against 0.632 without carryover). In the crossover design the closed form understates the mixed model's power (0.645 against 0.708 without carryover, and 0.279 against 0.31 to 0.35 under the constant rule). The mixed model's time term uses the between-path contrast that the paired difference discards. The derivation gives three results.

- **Proposition 1.** The variance of Δ does not depend on the carryover half-life. The half-life enters only through the response means and the correlation profile. Neither appears in the variance.
- **Proposition 2.** The population slope is

$$\beta^{\mathrm{E9}}_{bm:D} = -c_{bm}\frac{\sigma_{BR}}{\sigma_{bm}}\left(1-\bar{g}_{\mathrm{off}}\right), \qquad \beta^{\mathrm{E9b}}_{bm:D} = -c_{bm}\frac{\sigma_{BR}}{\sigma_{bm}}\left(1-\frac{n_{\mathrm{off}}}{n_{\mathrm{off}}+1}\,\bar{g}_{\mathrm{off}}\right),$$

  where $\bar{g}_{\mathrm{off}}$ is the mean of g_t over the n_{off} post-baseline off-drug visits. The slopes are specific to each path. In the Hybrid design n_{off} is 3 in two paths and 4 in the other two. The pooled slope is their allocation-weighted average.
- **Power** is a noncentral-t probability governed by the correlation $r_{\Delta B}$ between Δ and B. For E9 that correlation is proportional to the on/off correlation difference $\mathcal{G} = 1-\bar{g}_{\mathrm{off}}$. For E9b it is proportional to $1-\frac{n_{\mathrm{off}}}{n_{\mathrm{off}}+1}\bar{g}_{\mathrm{off}}$.

So carryover can affect power only through the on/off correlation difference. The propositions reduce the question of why power falls to the question of what the correlation rule does to $\bar{g}_{\mathrm{off}}$.

The feasibility bound

Write the response block of the correlation matrix as M and the biomarker's correlations with the response components as $c_{bm}\tilde{g}$, with $\tilde{g}$ the correlation profile padded with zeros at the TV and PB entries. The full matrix is positive definite if and only if M is, and

$$c_{bm}^2\,\tilde{g}^\top M^{-1}\tilde{g} < 1.$$

The left side is the squared multiple correlation of the biomarker on the whole response trajectory. The largest feasible c_{bm}, the feasibility bound, is therefore $c^*_{bm} = (\tilde{g}^\top M^{-1}\tilde{g})^{-1/2}$ (Appendix A.4). It

depends on the correlation structure, the design and the correlation profile. It is a bound on the DGP, not on any analysis.

Simulation studies

We designed and report the simulations following the ADEMP structure of Morris et al. (17,45): aims, data-generating mechanisms, estimands, methods and performance measures. There are three studies (Table 2). The scripts are listed under Availability of data and materials. The closed-form calculations of Stages 4 and 5 and of the planning section use the same code as study 1 with ρ varied. A fourth run, described in the planning section, simulates the analysis G10 of the companion paper (46) under the recommended DGP with the starting parameters. It supplies the G10 columns of the planning tables, the G10 sample sizes for 80 percent power and the G10 panel of Figure 4B (Figure 4). Its generator reproduces the closed-form power of E9 within two Monte Carlo SEs in the Hybrid, OL+BDC and crossover designs ($N = 70$, $c_{bm} = 0.25$, $t_{1/2} = 0.2$ week, 1,000 replicates; `42-check-param-sets.R`).

Table 2: Design of the three simulation studies in the ADEMP structure. MCSE: Monte Carlo standard error of a rejection rate, $\sqrt{p(1-p)/R}$ for R replicates.

	1. Closed-form verification	2. The starting code	3. Stress test under a prognostic biomarker
Aim	Check the closed-form moments and power of Appendix A, and the loss of power under the constant rule	Show on the starting code that the correlation rule alone determines the loss of power	Find which analyses keep their nominal size under features the recommended DGP adds

	1. Closed-form verification	2. The starting code	3. Stress test under a prognostic biomarker
Data-generating mechanisms	The correlation encoding with the constant rule and with the decaying correlation; compound symmetry and the separable AR(1) form; carryover on the drug-response mean decaying from the last on-drug value; Hybrid, crossover and OL+BDC designs; $N = 70$; grids of c_{bm} and $t_{1/2}$	The original `generateData()` at commit 58b32a9, as published and with the decaying correlation only (Appendix B.1); $c_{bm} = 0.25$. (a) Hybrid, $N = 70$, no censoring, $t_{1/2} \in \{0, 0.1, 0.2, 1\}$; (b) four designs, $N \in \{35, 70\}$, $t_{1/2} \in \{0, 0.1, 0.2\}$, the starting censoring patterns	The recommended DGP (Table 11), run three times: the reported run as defined, the first with the compounding carryover rule and the second with randomized allocation; base cell separable AR(1), $\rho = 0.7$, $N = 70$, $t_{1/2} = 0.5$ week; 15 null cells (11 without and 4 with a prognostic pathway of strength 0.3) and 26 alternative cells: power curves under the separable form (c_{bm} from -0.30 to 0.45, 7 cells) and under compound symmetry ($\rho_{cs} = 0.8$, c_{bm} from -0.20 to 0.25, 6 cells), and 13 cells at $c_{bm} = 0.30$ that each vary one feature (N, half-life, decay form, dropout, measurement error or a prognostic pathway)
Estimand	$\beta_{bm:D}$, and the mean of the paired-difference statistic	$\beta_{bm:D}$	$\beta_{bm:D}$

	1. Closed-form verification	2. The starting code	3. Stress test under a prognostic biomarker
Methods	Paired-difference statistics E9 and E9b, closed form and simulated; the binary-indicator mixed model fitted with `lmer`	The binary-indicator analysis pipeline (`lmer`, model-based test) on both versions	Fourteen analyses fitted to the same simulated trials, of which the ten in Table 8 are reported; two-sided tests at $\alpha = 0.05$
Performance measures	Mean, SD and power of the statistic; agreement as (simulated − closed form)/MCSE	Power; mean interaction estimate; number of repaired correlation matrices	Null rejection rate; power; failed fits
Replicates	5,000 per cell (MCSE of power at most 0.007); 500 to 1,000 for `lmer`	(a) 500 per cell (MCSE at most 0.022); (b) 250 per cell (at most 0.032)	1,000 per null cell (MCSE 0.0069 at a rate of 0.05), 500 per alternative cell (at most 0.022)

Number of replicates. Replicate counts were set by the precision each comparison needs. In study 3, 1,000 null replicates give a two-MCSE band of 0.036 to 0.064 for a single cell. Pooling 11 non-prognostic null cells gives the band of 0.046 to 0.054 used in Stage 6. In study 1, a cell is flagged when the simulated and closed-form values differ by more than 2 MCSE. Chance alone produces this in about 4.6% of cells, or about 26 of the 566 cell-variants (three designs, three correlation structures, two statistics); 16 were flagged. The cell-variants are not independent, since each uses its own draws but the same designs.

Random numbers. Every cell, or chunk of a cell, sets its own seed from a fixed rule: a function of the cell's parameters in study 1, a shared seed per half-life (study 2a) or per design, sample size and half-life (study 2b), and $20261007 + 1000 \times \text{cell} + \text{chunk}$ in study 3. Results therefore do not depend on the number of cores or the order in which chunks run. Both versions of the starting code in study 2 use the same seeds. All analyses in study 3 are fitted to the same trials. The comparisons are therefore paired. R's default Mersenne-Twister generator was used throughout.

Failed fits. In study 3, a fit that returned an error was recorded as missing. In the reported run,

failures numbered 0 to 10 per analysis over 28,000 trials, and at most 0.3% in any cell. In the first and second runs they numbered 0 to 12 and 0 to 7, and at most 0.6% and 0.4% in any cell. In study 2 the starting pipeline returned an estimate for every replicate. No correlation matrix was repaired in either version, because $c_{bm} = 0.25$ lies below the compound-symmetry feasibility bound of 0.256. Study 1 does not fit models for the paired-difference statistic, so it has no failures.

Which form of the DGP each study used. Study 1 and the closed-form calculations build the cross-component correlation as $c_1 \rho^{|w_t - w_s|}$ (the separable form) where the separable structure is named, and as the starting constants where compound symmetry is named. Study 2 is the starting code and so the starting structure. Study 3 was run three times, with the separable form in every cell except the compound-symmetry curve and an analysis half-life of 0.5 week. The first run used fixed path allocation (18, 18, 17, 17) and the compounding carryover rule on the mean. The second run used carryover decaying from the last on-drug mean, the rule of the recommended DGP, and also assigned each participant to a path at random in every trial. Randomized allocation is a driver option, not part of the recommended DGP as defined (Table 11). The third run is the recommended DGP as defined: the anchored carryover rule with fixed allocation. It differs from the first run only in the carryover rule and from the second only in the allocation, so the two effects can be separated. The tables in Stage 6 are from the third run. The first run is in Appendix C, Table 17.

Software. R 4.6.1, with lme4 2.0.1 and lmerTest 3.2.1 for the starting pipeline, nlme 3.1-169, mmrm 0.3.17, clubSandwich 0.7.0 and MASS 7.3-65. In study 3 the binary-indicator analysis is fitted with `mmrm` and a compound-symmetry covariance with Satterthwaite inference, not with `lmer`. The two coincide when the within-person correlation is non-negative.

Exploratory status. The stress test is exploratory. Its cells and analyses were written down as a working protocol (Additional file 2). No analysis plan was fixed in advance, and the protocol was revised as results came in. Two cells with a prognostic association between the biomarker and baseline severity, and three analyses, were added after the first results. Failed fits were first counted as non-rejections and are now excluded from the denominator, a change of at most 0.0025 in any rate. The study also compared the analyses' power and selected one by a decision rule. That selection was revised after results, is post hoc, and is reported in Additional file 3, not here. Such post hoc changes can favor one method in comparative simulation studies (47), which is a further reason not to rank the analyses here. This paper uses the stress test only for what it shows about the DGP: which analyses are valid under a prognostic biomarker, and what the optional features

do to size and power.

Results

The Results are organized as stages. Each stage takes one element of the DGP. It states the candidate choices and the evidence. It ends with the choice made and the criterion it rests on. Table 11, at the end, collects the choices.

Stage 1. Feasibility of the joint distribution

The first question is whether the joint distribution exists for the effect sizes a study simulates. Under compound symmetry the correlation matrix is positive definite only for c_{bm} below a bound. Without carryover the bound is 0.256 in the Hybrid design. It is 0.256 in the crossover and 0.262 in the open-label-plus-blinded-discontinuation design. These values follow from the feasibility bound with M the starting response block.

But why is the bound so low? It is the condition that the biomarker's squared multiple correlation on the whole response trajectory stays below one. Under compound symmetry every visit's drug response shares the same person-level factor. The biomarker therefore cannot be correlated with the on-drug visits alone. Through that factor it is correlated with the off-drug visits as well. The bound is low for two reasons. The correlation profile switches between visits that the structure ties together. The drug-response component also keeps its full variance at off-drug visits. Stages 4 and 5 show how far a serial correlation that decays with lag raises the bound. Even then, an effect size of 0.6 is infeasible in 75% of the extended grid and in every Hybrid cell (Appendix B.2, Table 16).

When a matrix is infeasible, a DGP can either repair it or refuse it. The starting code repairs it without a warning (Appendix B.2). A repaired matrix simulates a different DGP from the one stated, and it does not report which correlation changed. The recommended DGP therefore stops and reports the cell as infeasible. The package function `buildSigma()` does so by default (Appendix B.5). The choice rests on the existence criterion. What the bound means in outcome units, and what to do above it, are taken up in the planning section and the Discussion.

Stage 2. The biomarker-response correlation under carryover

This stage takes the correlation at off-drug visits. Power to detect the interaction falls sharply under the constant rule once carryover is introduced (3). In our 500-replicate rerun of the starting code in the Hybrid design at $N = 70$ and $c_{bm} = 0.25$ (Table 4), power fell from 0.668 without carryover to 0.126 at a half-life of 0.1 week. A 4,000-replicate rerun of that cell gives 0.090 (below). No correlation matrix was repaired in that run, because 0.25 lies below the compound-symmetry feasibility bound of 0.256. At that half-life only 0.1% of the drug effect remains a week after stopping.

Under the constant rule every off-drug visit in the Hybrid design follows an exposure. With any positive half-life its drug-response mean is positive, so $g_t = 1$. Hence $\bar{g}_{\text{off}} = 1$ and, by Proposition 2,

$$\beta^{\text{E9}}_{bm:D} = 0, \qquad \beta^{\text{E9b}}_{bm:D} = -c_{bm}\frac{\sigma_{BR}}{\sigma_{bm}}\frac{1}{n_{\text{off}}+1} \qquad \text{for every } t_{1/2} > 0.$$

Power is a step function of the half-life (Table 3, Figure 1). It takes its no-carryover value at $t_{1/2} = 0$. It takes a constant value at every positive half-life of practical interest, whether 0.1 week or 10. The exception below about 0.007 week is an artifact of double-precision arithmetic in the starting code, not a property of the DGP (Appendix B.1). For E9b the constant is 0.081. This value comes from the baseline row, which alone keeps $g_t = 0$.

The binary-indicator analysis behaves the same way. In our reconstruction of the starting DGP, the mean estimate falls from −0.126 to between −0.026 and −0.032 at every positive half-life. Power falls from 0.63 to between 0.08 and 0.11 (500 replicates per cell). The separately seeded run of the starting code reported above gives 0.126 at 0.1 week, against 0.080 for the reconstruction. The difference is 2.4 combined Monte Carlo SE (two-sided $p = 0.016$). The two implementations share the parameters, the path allocation and the analysis model. They differ in the random-number stream and in the carryover rule on the mean. At 0.1 week the carryover rule leaves an off-drug drug-response mean below 0.011 points under either rule, so it cannot account for a difference of this size. We therefore reran the starting code at 0.1 week with 4,000 new replicates (16 seeds not used before). Power was 0.090 (MCSE 0.0045), within 0.8 combined Monte Carlo SE of the reconstruction. Rerunning the stored cell with its stored seed reproduces 0.126 exactly. The rerun is therefore consistent with the reconstruction, and the value of 0.126 is consistent with a high draw of a 500-replicate run. We found no difference between the two implementations that would explain it.

Table 3: Closed-form power by carryover half-life (weeks). Hybrid design, $N = 70$, $c_{bm} = 0.25$, compound symmetry. Constant: the constant rule (constant correlation after first exposure). Decaying: the decaying rule (correlation decaying at the carryover half-life). E9: paired-difference statistic on post-baseline visits; E9b: the same with baseline counted as an off-drug visit, as in the binary-indicator analysis. Against Monte Carlo estimates (5,000 replicates per cell), 2 of the 24 cell-variants in this table differ by more than 2 Monte Carlo SE, and 16 of all 566 cell-variants of study 1 (about 26 expected by chance).

Correlation rule	Statistic	0	0.1	0.2	0.5	1	2
Constant	E9	0.721	0.050	0.050	0.050	0.050	0.050
	E9b	0.654	0.081	0.081	0.081	0.081	0.081
Decaying	E9	0.721	0.721	0.712	0.631	0.473	0.278
	E9b	0.654	0.653	0.647	0.586	0.472	0.327

With a constant correlation, the biomarker moderates the on-drug and post-baseline off-drug responses equally. No interaction remains among the post-baseline visits. Only the baseline visit, where the correlation is zero, still contrasts with the on-drug visits, which is why E9b and the binary-indicator analysis keep a little power. The loss of power under small carryover is therefore a property of the correlation rule, not of carryover or of the designs.

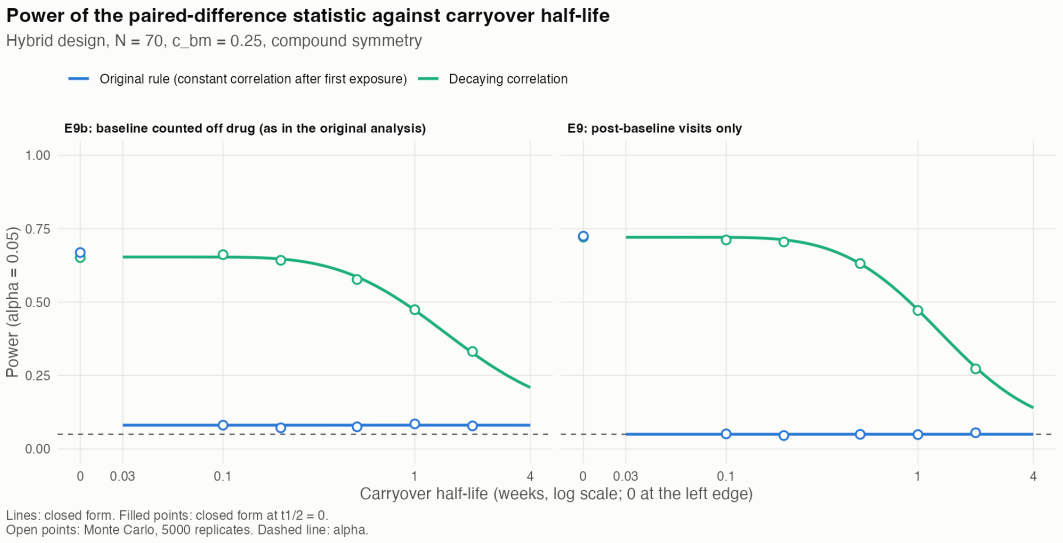


Figure 1: Closed-form power (lines) against carryover half-life, with Monte Carlo estimates (points; 5,000 replicates per cell), under the constant rule (constant correlation after first exposure) and the decaying rule (correlation decaying at the carryover half-life). Hybrid design, $N = 70$, $c_{bm} = 0.25$, compound symmetry; left panel E9b, right panel E9.

The candidate is a correlation that decays after discontinuation at the carryover half-life: $g_t = 1$ on drug, $2^{-t_{sd,t}/t_{1/2}}$ off drug after exposure, and 0 before any exposure. This is the pharmacological reading of a predictive biomarker. The biomarker predicts the size of the drug effect, so it cannot

predict a drug effect that is no longer present. Under the decaying rule the mean of g_t over the off-drug visits is

$$\bar{g}_{\text{off}}(t_{1/2}) = \frac{1}{n_{\text{off}}} \sum_{t\,\text{off}} 2^{-t_{sd,t}/t_{1/2}},$$

which is strictly increasing in $t_{1/2}$, because every off-drug visit has $t_{sd} \geq 1$ week. Two consequences follow.

- **Power is strictly decreasing in the half-life** for the path-stratified paired-difference statistic. This holds on every path with at least one off-drug visit after an exposure (Proposition 3, Appendix A.3). The placebo-first crossover path, whose off-drug visits all precede exposure, has constant power. For the pooled statistic power is non-increasing at every step of a 120-point grid from 0.03 to 4 weeks. For the binary-indicator mixed model only the simulated half-lives are available, and its values there are consistent with a monotone decline within Monte Carlo error (Table 4).
- **The loss vanishes faster than any power of the half-life as $t_{1/2} \to 0$.** At 0.1 week, $\bar{g}_{\text{off}} \leq 2^{-10}$ and power is unchanged to within 0.001.

The constant rule is the limit of the decaying rule as $t_{1/2} \to \infty$, applied at every positive half-life.

Power depends on the correlation profile only through the on/off correlation difference $\mathcal{G} = 1 - \bar{g}_{\text{off}}$ (Figure 2). The difference is 1 without carryover. Under the decaying rule in the Hybrid design it is 0.9997 at 0.1 week, 0.991 at 0.2, 0.906 at 0.5, 0.754 at 1 and 0.551 at 2. Under the constant rule it is 0 at every positive half-life. The noncentrality is proportional to $r_{\Delta B}/\sqrt{1 - r_{\Delta B}^2}$, and $r_{\Delta B}$ is proportional to the difference. For small $r_{\Delta B}$ the sample size needed for a given power therefore scales approximately as $1/\mathcal{G}^2$. At 1 week this is about 1.8 times the no-carryover sample. At 2 weeks it is about 3.3 times.

Replacing only the correlation lines of the starting code with the decaying rule (Appendix B.1) and running the pipeline unchanged gave the results in Table 4. The two versions coincide without carryover, as they must. At 0.1 and 0.2 week the decaying version keeps the full interaction, within Monte Carlo error of the no-carryover value of 0.668. At 1 week it shows a gradual decline. No matrix was repaired in either version, because $c_{bm} = 0.25$ is below the feasibility bound.

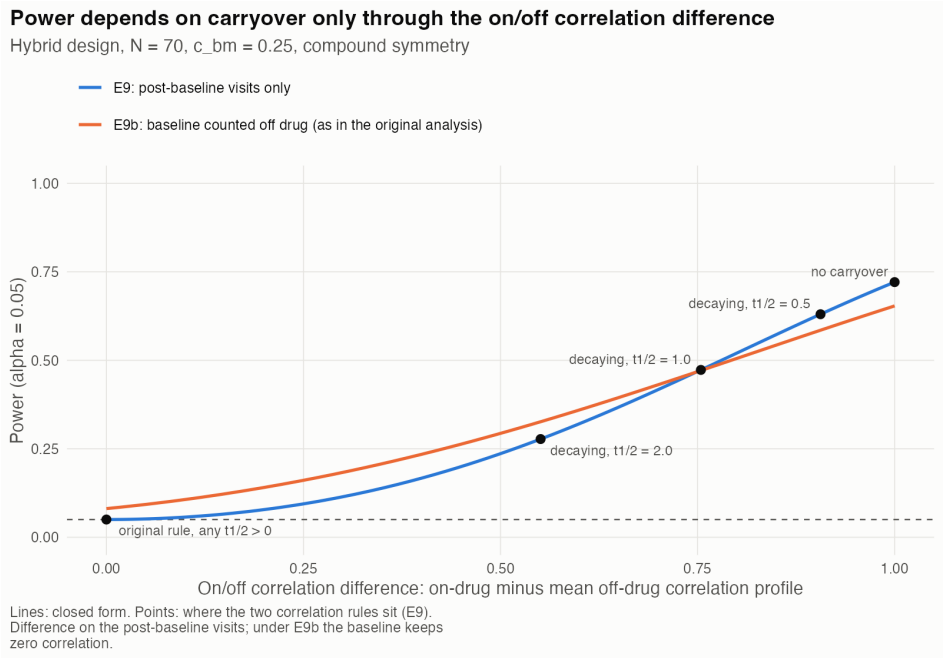


Figure 2: Closed-form power against the on/off correlation difference, with the positions of the two correlation rules. Hybrid design, $N = 70$, $c_{bm} = 0.25$, E9.

Table 4: The starting code at commit 58b32a9, as published (constant) and with the decaying correlation (decaying): Hybrid design, $N = 70$, $c_{bm} = 0.25$, 500 replicates per cell, same seeds. Monte Carlo SE of power at most 0.022. A rerun of the original version at 0.1 week with 4,000 new replicates gave power 0.090 (Stage 2 text).

$t_{1/2}$ (weeks)	Mean estimate, constant	Mean estimate, decaying	Power, constant	Power, decaying
0	-0.131	-0.131	0.668	0.668
0.1	-0.028	-0.129	0.126	0.638
0.2	-0.029	-0.131	0.092	0.656
1	-0.033	-0.107	0.094	0.462

Figure 3 repeats the layout of Figure 4B of the earlier study on the starting code, for all four designs, both sample sizes and the five censoring patterns, as published and with the decaying correlation. It uses $c_{bm} = 0.25$ instead of the earlier 0.6. The value 0.6 lies above the feasibility bound of the Hybrid design and of most cells of the other designs, so it would confound the comparison with the repair. At 0.25 neither version repaired a matrix, so the correlation rule is the only difference. With 250 replicates per cell the Monte Carlo SE reaches 0.032, so a difference of 0.05 between adjacent cells is within Monte Carlo error. Only the pattern across cells should be read. Results at $N = 70$ without censoring:

- **Hybrid design.** Power under the constant rule falls from 0.66 to 0.08 at both positive half-lives. Power under the decaying rule stays at 0.62 and 0.68.
- **Open label plus blinded discontinuation.** Power under the constant rule falls from 0.57 to 0.12 and 0.13. Power under the decaying rule stays at 0.57 and 0.60.
- **Crossover.** Power under the constant rule falls only to 0.34 and 0.33. The placebo-first path's off-drug visits precede any exposure and so keep zero correlation under both rules. Power under the decaying rule stays at 0.71 and 0.78.
- **Open label.** There are no off-drug visits. The two rules coincide and power is low throughout.

The same pattern holds at $N = 35$ and under every censoring pattern.

Figure 4 repeats the grid of Figure 3 under the recommended DGP of Stage 7 with the analysis G10 that the companion paper recommends (46). G10 is a random-intercept model with CAR(1) residual correlation and an exposure-decayed drug indicator whose half-life is chosen by AIC on the model without the interaction, with CR2 cluster-robust standard errors (planning section). Both figures use the starting response and baseline parameters (Methods). The data-generating process differs from that of Figure 3 in its correlation structure (separable AR(1) at $\rho = 0.7$ instead of compound symmetry) and in its carryover rule on the mean, and the analysis differs, so the two figures are not matched cell by cell. Each cell has 250 replicates and a Monte Carlo SE of at most 0.031. The open-label design is not estimable under G10: every visit is on drug, so the exposure-decayed indicator is constant at 1 and the interaction term is collinear with the biomarker. The binary-indicator analysis replaces the interaction with a biomarker-by-time term in that case. G10 has no such substitute, and the cells are marked as not estimable rather than simulated. The censoring patterns in this figure use the corrected dropout rule. The starting `censordata()` raised

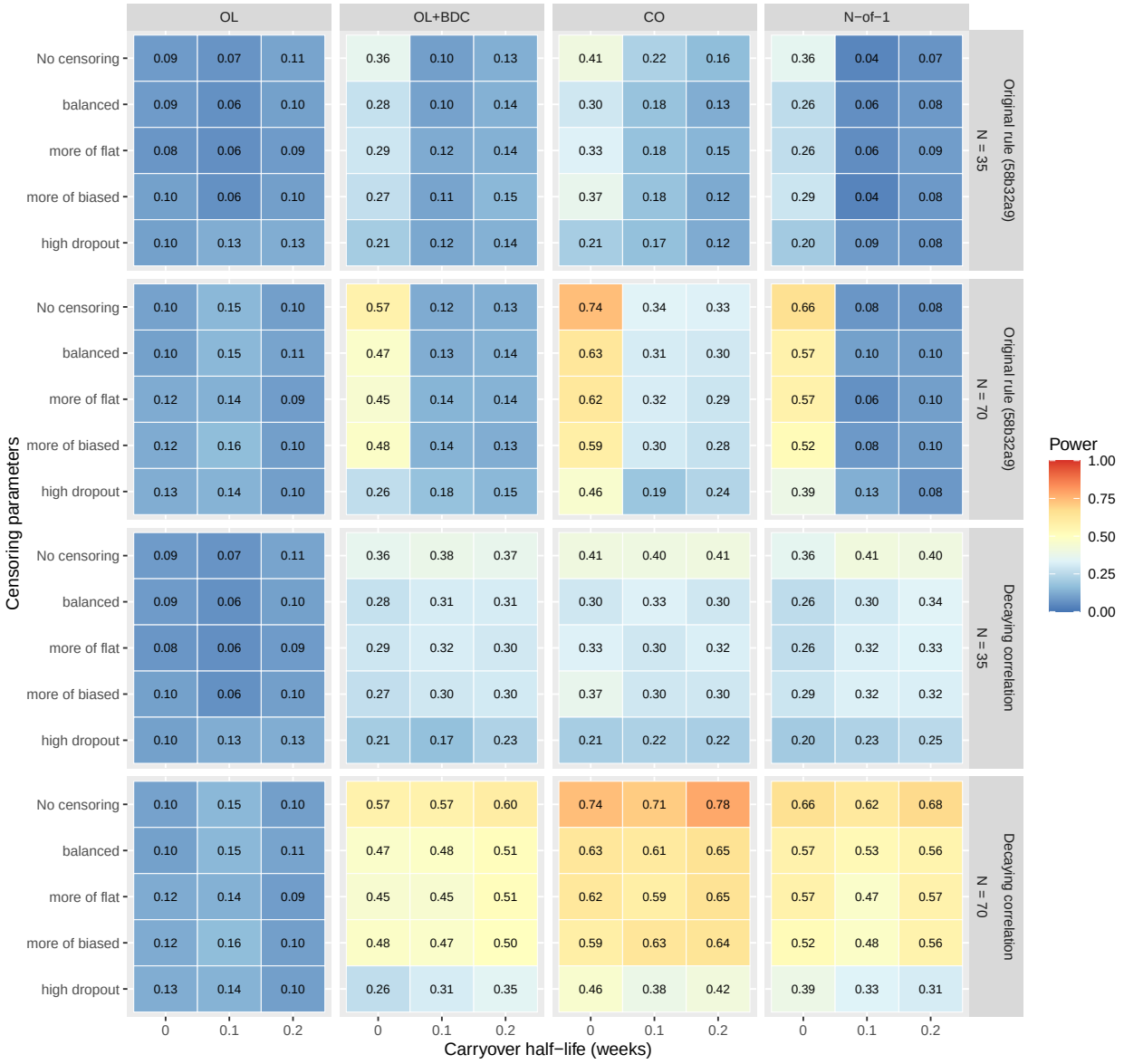


Figure 3: Power to detect the biomarker-by-treatment interaction, in the layout of Figure 4B of the earlier study, from the starting code at commit 58b32a9 as published (upper two panel rows) and with the decaying correlation (lower two panel rows). Rows within panels: censoring patterns; columns: carryover half-life; panels: trial design (N-of-1 is the Hybrid design) and total sample size. $c_{bm} = 0.25$, 250 replicates per cell (Monte Carlo SE at most 0.032), the same seeds for both versions. No matrix was repaired in either version.

dropout with improvement, the reverse of the intended mechanism, through a sign error in its biased term (Appendix B.5). Under the corrected rule, dropout rises with worsening. Figure 3 keeps the published rule, because it reproduces the starting code. Results at $N = 70$ without censoring:

- **Hybrid design.** Power is 0.34, 0.38 and 0.34 at half-lives of 0, 0.1 and 0.2 week.
- **Open label plus blinded discontinuation.** Power is 0.34, 0.28 and 0.24.
- **Crossover.** Power is 0.20, 0.19 and 0.27.

We compare G10 power at the pharmacological half-lives with its no-carryover value. Two rates differ when they are more than two combined Monte Carlo SEs apart. In the Hybrid design, power is within Monte Carlo error of its no-carryover value at both half-lives. In the OL+BDC design it is within Monte Carlo error of its no-carryover value at 0.1 week and lower (0.24) at 0.2 week. In the crossover design it is within Monte Carlo error of its no-carryover value at both half-lives. Each comparison rests on one cell of 250 replicates. In the planning run of the same DGP and analysis, with 1,000 replicates per cell (Table 13), OL+BDC power is 0.295 without carryover and 0.275 at 0.2 week, within Monte Carlo error of each other. Of the censoring patterns, the 'high dropout' pattern gives the lowest power in the Hybrid design at 0.2 week, 0.14.

G10 power is lower than in the starting code with the decaying correlation (Figure 3, lower two panel rows) in every design. Without carryover it is 0.34, 0.34 and 0.20 in the Hybrid, OL+BDC and crossover designs, against 0.66, 0.57 and 0.74 in the starting code. The analysis does not account for the difference. Under the recommended DGP the closed-form power of the paired-difference statistic E9 at the same N and c_{bm} is 0.29, 0.21 and 0.20, computed as for Table 12, no higher than G10. Without carryover the two carryover rules on the mean coincide, so the difference comes from the correlation structure. The starting code uses compound symmetry at $\rho_{cs} = 0.8$, and the recommended DGP uses the separable AR(1) form at $\rho = 0.7$. Table 5 shows the effect of that change on closed-form E9 power in the Hybrid design. The drop is largest in the crossover design. We have not established why.

Figure 5 extends the comparison over interaction strengths and longer half-lives. It shows closed-form power in the Hybrid design for each correlation rule.

- **The constant rule.** Every column after $t_{1/2} = 0$ falls to the test size or just above it, at every strength. For example, at $c_{bm} = 0.2$ power under E9b falls from 0.46 to 0.07.

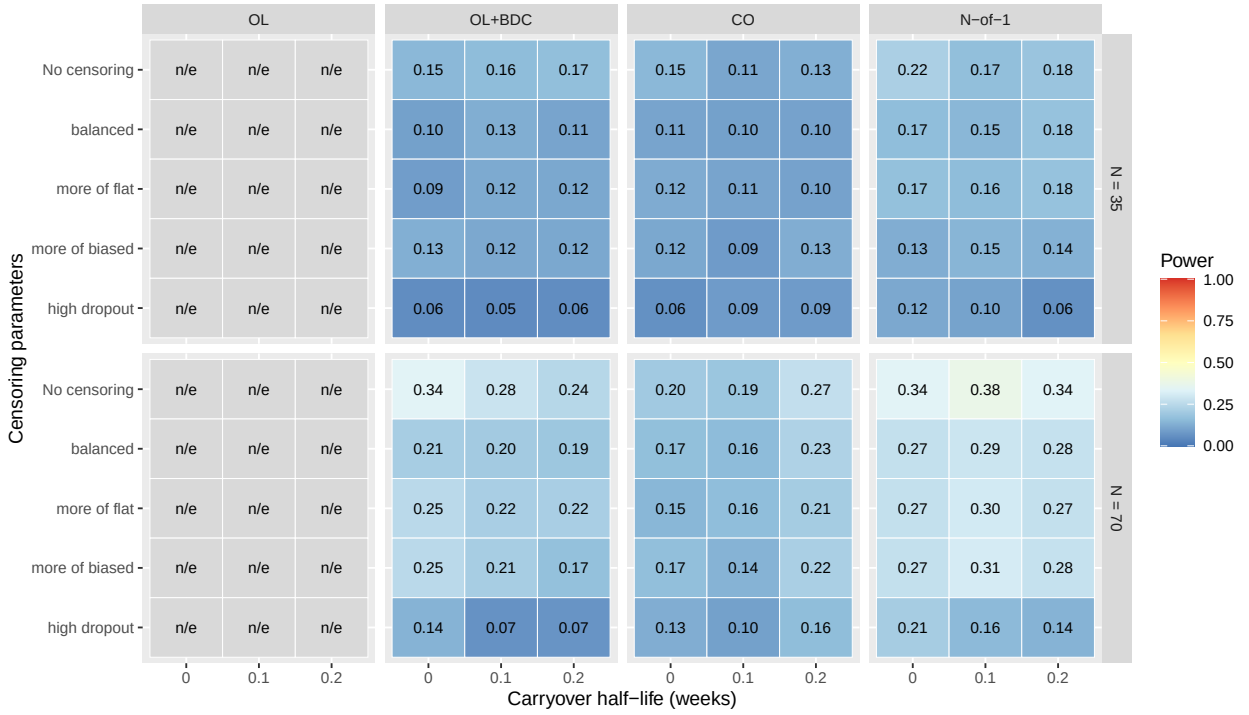


Figure 4: Power of the analysis G10 to detect the biomarker-by-treatment interaction under the recommended DGP (separable AR(1), $\rho = 0.7$, $c_1 = 0.2$, decaying correlation, carryover decaying from the last on-drug mean, original response parameters), in the layout of Figure 4B of the original study. Rows within panels: censoring patterns; columns: carryover half-life; panels: trial design (N-of-1 is the Hybrid design) and total sample size. $c_{bm} = 0.25$, 250 replicates per cell (Monte Carlo SE at most 0.031). Censoring patterns use the corrected dropout rule, under which dropout rises with worsening; the original-code figure uses the published rule of commit 58b32a9. Gray cells (n/e): the open-label design, in which every visit is on drug and the interaction is not estimable under G10.

- **The decaying rule.** Power declines smoothly: 0.46, 0.46, 0.46, 0.41, 0.32 and 0.23 at the same strength over half-lives of 0 to 2 weeks.
- **Feasibility.** Gray cells mark effect sizes at or above the compound-symmetry feasibility bound, where no joint distribution exists. Under the decaying rule, $c_{bm} = 0.3$ becomes feasible only from $t_{1/2} = 1$ week. A correlation that decays instead of switching places a smaller demand on the structure (Appendix A.5).

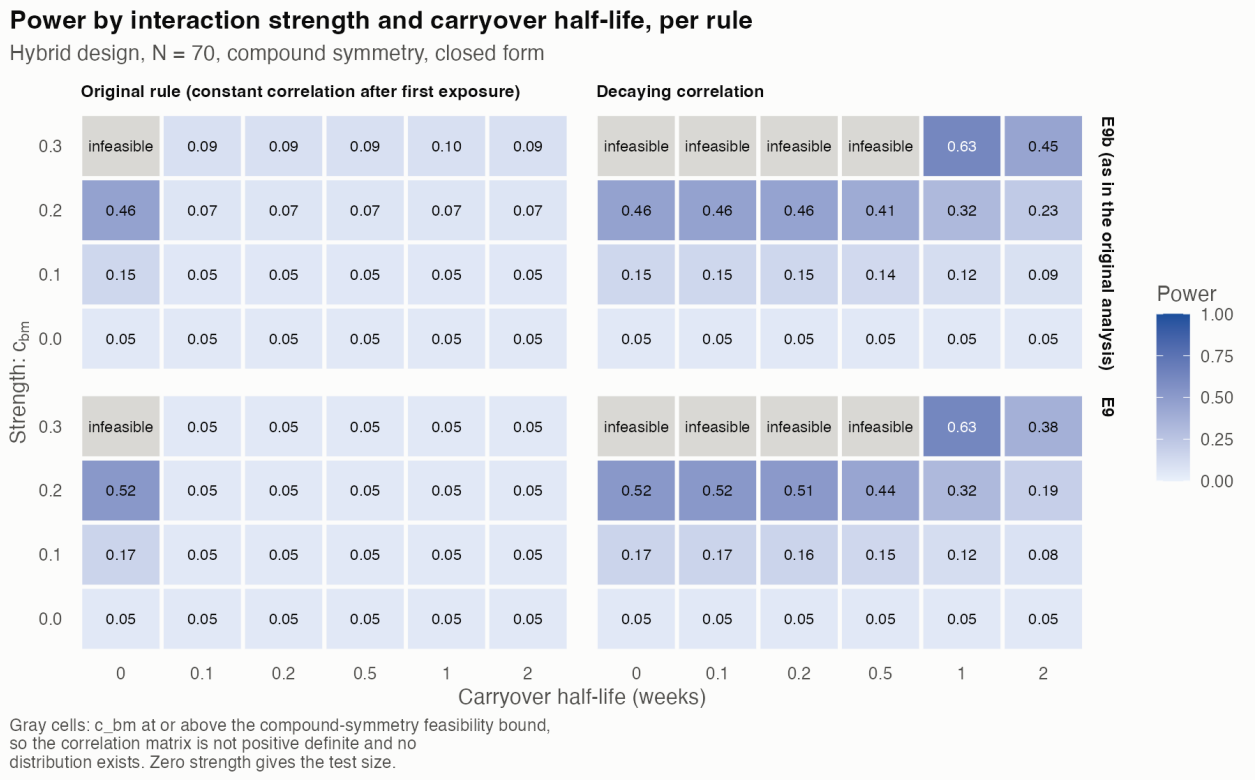


Figure 5: Closed-form power of the paired-difference statistic by interaction strength (c_{bm}) and carryover half-life, in the layout of Figure 4B of the earlier study: the constant rule and the decaying correlation. Upper row: E9b, which counts baseline as an off-drug visit as the binary-indicator analysis does; lower row: E9. Gray cells: c_{bm} at or above the compound-symmetry feasibility bound, where the correlation matrix is not positive definite. Hybrid design, $N = 70$, compound symmetry.

The recommended DGP uses the decaying correlation. The choice rests on identifiability under carryover and on fidelity to the pharmacology. Two limitations of the choice are stated here. First, at the pharmacological half-lives of 0.1 to 0.2 week the on/off correlation difference is within 0.01 of 1. The decaying rule therefore returns the no-carryover result. The recommended DGP at its default half-life is not a test bed for carryover. The smooth decline is a statement about half-lives of 0.5 week and above, which we label behavioral. Second, the decaying correlation is asymmetric.

Off drug the moderation fades with the drug effect. On drug it is the full $c_{bm}\sigma_{BR}b_i$ at every visit, including the first weeks of titration when the Gompertz mean is a small fraction of its maximum. A correlation proportional to the current drug-response mean at every visit, or a correlation half-life separate from the mean's, would remove the asymmetry. Neither was tried. The sensitivity model of the Discussion is the form that addresses it. The mixed-model power of the recommended DGP at half-lives of 0.5 to 2 weeks, with the half-life fixed in the analysis or chosen from the data, is the subject of a companion paper (46).

Stage 3. Carryover on the mean

The compounding carryover rule decays the previous visit's mean by the total time since discontinuation. At the second and later off-drug visits that mean has already been decayed, so the decay compounds. Consider two off-drug visits at weeks 11 and 12 following discontinuation at week 10. The starting code decays the week-10 value by $2^{-1/t_{1/2}}$ and then by $2^{-2/t_{1/2}}$, a total exponent of 3 weeks for 2 weeks elapsed. We treat the two forms as two specifications.

The recommended DGP decays the mean from the last on-drug value, $\mu_{BR,\text{last}}\,2^{-t_{sd}/t_{1/2}}$. The mean at any off-drug visit then depends only on the elapsed time (Appendix B.3).

The choice changes no slope, variance or feasibility bound. It changes only the path means. It therefore changes the closed-form power of study 1 by at most 0.002 in the Hybrid design. In the starting pipeline it changes power by at most 0.006. Figure 6 shows the drug-response mean by visit for the four Hybrid paths under both forms. Two features are visible. At $t_{1/2} = 1$ week the constant rule lies below the decaying rule at the second off-drug visit, as expected. At 0.2 week both are indistinguishable from zero one week after discontinuation. At re-exposure (week 16 in paths 1 and 3) time on drug restarts at zero. The mean then rises along a fresh Gompertz curve to 4.3 points at the week-16 visit, four weeks into the restarted curve. The residual from the earlier exposure has decayed to 0.16 points by then even at a 1-week half-life. No downward step therefore appears on the Hybrid schedule. A step would appear only if a visit fell within about a week of re-exposure at a half-life of a week or more.

The recommended DGP decays carryover from the last on-drug mean. It keeps the starting DGP's restart of time on drug at re-exposure, which for prazosin matches re-titration after an interruption. For drugs without titration the response should instead rebuild from its residual level. That is

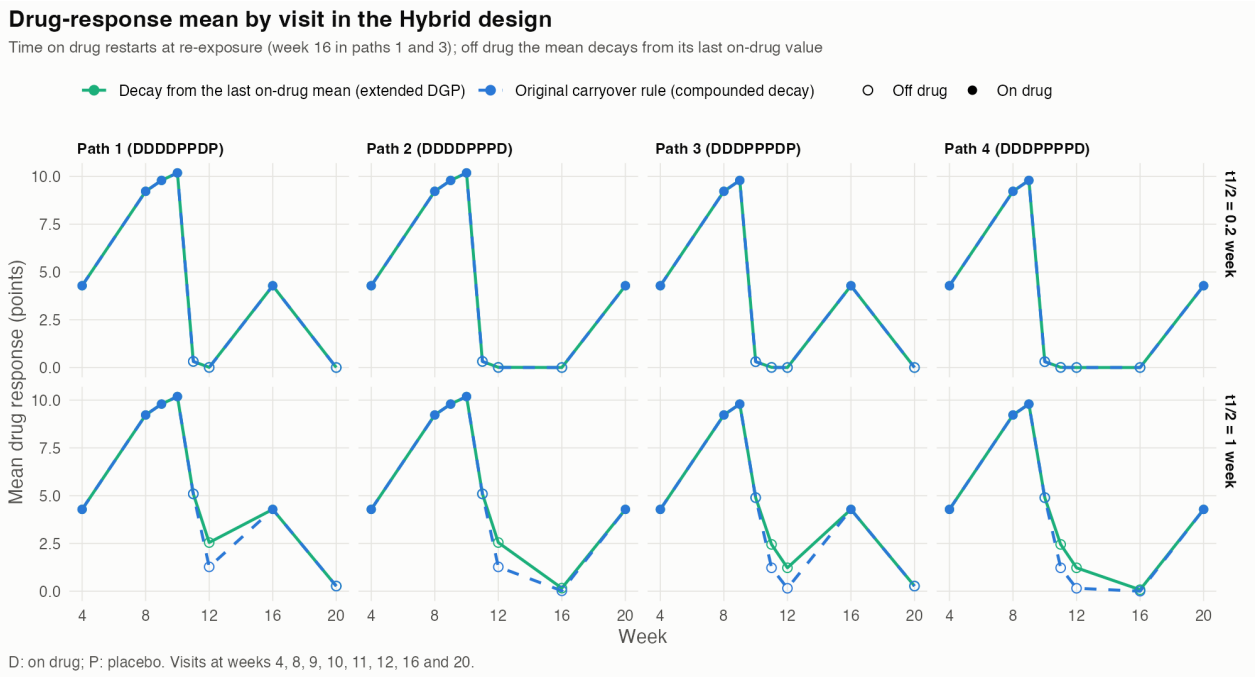


Figure 6: Drug-response mean by visit in the Hybrid design, by path, under the anchored carryover rule (decay from the last on-drug mean, the recommended DGP) and the compounding carryover rule (compounded decay), at carryover half-lives of 0.2 and 1 week. Filled points: on-drug visits; open points: off-drug visits. D: drug; P: placebo.

an option, not the default. The choice rests on fidelity to the pharmacology, as a judgment. The numerical consequence is at most 0.006 in power either way.

Stage 4. Serial correlation

The starting DGP uses compound symmetry. Visits one week apart and sixteen weeks apart have the same correlation of 0.8. The one PTSD symptom series available to us, the total CAPS score in CSP 563 (planning section), has a correlation that decays slowly with lag. Matching its 4- and 20-week correlations within this DGP's structure would need ρ of about 0.87 to 0.91 per week (Additional file 1). That series is a total score, so it constrains the component-level ρ without determining it. Both compound symmetry and AR(1) are used in N-of-1 simulation (19). Serial correlation has been discussed as a design consideration for personalized studies (48). Two elements are open: the form of the within-component correlation, and its value.

The recommended DGP uses AR(1) in calendar weeks, $\rho^{|w_t - w_s|}$, with one ρ for all three components. The value of ρ is a parameter. We take $\rho = 0.7$ as a convention only, not an estimate. Under the adopted separable form an effect size of 0.3 is already feasible at $\rho = 0.8$ (bound 0.440), so feasibility

Table 5: Serial correlation: the feasibility bound on c_{bm} (Ceiling; decaying correlation, no carryover, minimum over the Hybrid, crossover and OL+BDC designs) and closed-form E9 power (Hybrid, $N = 70$, no carryover) by ρ, under compound symmetry and under the separable AR(1) form of Stage 5. Under compound symmetry the Hybrid and crossover designs tie for the minimum. Under the separable form the minimum is attained in the Hybrid design at $\rho = 0.8$ and in the OL+BDC design at $\rho \leq 0.7$, and the crossover design is never binding. At $t_{1/2} = 0.2$ week every power value is lower by 0.004 to 0.009.

	Compound symmetry		Separable AR(1)	
ρ_{cs} or ρ	Ceiling	Power at $c_{bm} = 0.25$	Ceiling	Power at $c_{bm} = 0.25$
0.5	0.458	0.492	0.448	0.261
0.6	0.410	0.553	0.462	0.270
0.7	0.346	0.628	0.474	0.290
0.8	0.256	0.721	0.440	0.335

at 0.3 does not determine the value. Of the four values in Table 5, 0.7 has the largest separable feasibility bound (0.474). That keeps the planning effect sizes feasible but says nothing about whether 0.7 is right, and the value lies below the range the CSP 563 series suggests. Above 0.7, a larger ρ lowers the feasibility bound and raises E9 power (Table 5, 0.7 against 0.8). If that continues into the range CSP 563 suggests, the planning results at $\rho = 0.7$ overstate the largest feasible effect and the sample size needed. Table 6 extends the separable form to ρ from 0.4 to 0.95. The bound is largest at $\rho = 0.7$. It falls slowly below that value, to 0.437 at 0.4, and steeply above it, to 0.340 at 0.9 and 0.247 at 0.95, where the Hybrid design is the binding one. Above 0.7 the E9 sample size in the Hybrid design falls as well. The planning section repeats its calculations at $\rho = 0.5$ and 0.9.

Table 5 shows what ρ does under compound symmetry and under the structure adopted in Stage 5, for $\rho \in \{0.5, 0.6, 0.7, 0.8\}$. Under compound symmetry alone, lowering ρ_{cs} from 0.8 to 0.7 raises the feasibility bound from 0.256 to 0.346. An effect size of 0.3 then becomes feasible by the change of ρ alone, with no change of structure. The same change lowers closed-form power at $c_{bm} = 0.25$ from 0.721 to 0.628. A lower serial correlation leaves more within-person noise in the on-minus-off contrast. The two criteria pull in opposite directions. The feasible range widens as ρ falls, and the power at a matched c_{bm} falls.

The recommended DGP uses AR(1) in weeks at $\rho = 0.7$. The choice rests on existence, traded against power. The calibration status is open. The AR(1) form is adopted for the reason given in Stage 5 (it is what the analysis model assumes), not because compound symmetry has been shown wrong for these data. Any study using the DGP should report ρ. Where the conclusion could depend on it, the analysis should be repeated at other values, as the planning section does at 0.5

Table 6: Serial correlation under the separable AR(1) form of the recommended DGP ($c_1 = 0.2$, decaying correlation, carryover decaying from the last on-drug mean): the feasibility bound on c_{bm}, minimum over the Hybrid, crossover and OL+BDC designs, by carryover half-life, the design that attains the minimum without carryover, and the closed-form total N for 80 percent power of E9 in the Hybrid design at the 0.2-week half-life. Not feasible: c_{bm} lies above the Hybrid bound.

	Feasibility bound				Hybrid N, E9	
ρ	$t_{1/2} = 0$	0.2 week	1 week	Binding design	$c_{bm} = 0.25$	$c_{bm} = 0.35$
0.40	0.437	0.438	0.445	OL+BDC	318	160
0.50	0.448	0.450	0.465	OL+BDC	314	158
0.60	0.462	0.465	0.494	OL+BDC	302	152
0.65	0.469	0.473	0.512	OL+BDC	292	148
0.70	0.474	0.479	0.533	OL+BDC	278	140
0.75	0.465	0.471	0.550	Hybrid	258	130
0.80	0.440	0.446	0.544	Hybrid	234	118
0.85	0.400	0.407	0.519	Hybrid	200	100
0.90	0.340	0.346	0.457	Hybrid	154	not feasible
0.95	0.247	0.251	0.343	Hybrid	92	not feasible

and 0.9.

Stage 5. Cross-component structure

The starting DGP has a same-visit cross-component correlation $c_1 = 0.2$ and an other-visit correlation $c_\times = 0.1$ at every lag. With AR(1) within components, the direct translation keeps c_1 at lag zero and lets the other-visit correlation decay, $c_\times \rho^{|w_t - w_s|}$. This two-constant form is what the package's `buildSigma()` builds when its two constants differ (Appendix B.5). It is not the only choice, and it has a validity condition.

The recommended DGP uses the separable form, $c_1 \rho^{|w_t - w_s|}$ at every lag including zero. The response block is then a Kronecker product $M = K \otimes A$ of a 3×3 component correlation K (1 on the diagonal, c_1 off it) and one AR(1) kernel $A_{ts} = \rho^{|w_t - w_s|}$. In words, every component has the same first-order autoregressive correlation, and the between-component correlation is constant at every lag. It is the two-constant form with $c_\times = c_1$. The same-visit value of the starting DGP, $c_1 = 0.2$, is kept. Four properties follow (Appendix A.5).

1. **Validity for every schedule.** The eigenvalues of M are products of those of K and A, so M is positive definite for any visit times. The two-constant form adds a same-visit cross-component term that carries correlation without variance. It is positive definite only when

the smallest eigenvalue of A exceeds $(c_1 - c_\times)/(1 - c_\times) = 0.111$. On the paper's schedules the condition holds (smallest eigenvalue 0.122 in the Hybrid and OL+BDC designs at $\rho = 0.8$, 0.193 at 0.7; 0.28 and 0.43 in the crossover design). On 20 weekly visits at $\rho = 0.8$ it holds by less than 0.001. On daily measurement over eight weeks it fails (0.016 at $\rho = 0.8$, 0.026 at 0.7). The two-constant form therefore cannot be used for the dense schedules that wearable or diary outcomes produce.

2. **The starting structure is already of this kind in pieces.** It is the sum of a person-level and an occasion-level separable term, each valid. This is why its matrices fail only through the biomarker row, never in the response block.
3. **An exact, interpretable feasibility bound.** The bound factors into a component term and a schedule term,

$$c_{bm}^* = \sqrt{1 - R^2}\,\left(g^\top A^{-1} g\right)^{-1/2}, \qquad R^2 = \frac{2c_1^2}{1 + c_1},$$

where R^2 is the squared multiple correlation of the drug response on the other two components at one visit. The schedule term is a sum of per-step terms of the correlation profile. A profile that switches abruptly between closely spaced visits has a large per-step term. This is why designs with many on-off switches can carry only small effects.
4. **It nearly matches the analysis model.** Under separability the summed response has covariance $(s_t^\top K s_s)\,\rho^{|w_t - w_s|}$, with s_t the component SDs at visit t (Appendix A.5). With constant SDs this is exactly AR(1), the residual structure `corCAR1` assumes. In the Hybrid design the placebo-response SD is $10\eta_t$, so it halves at blinding. The summed-response variance is then 368 at open-label visits and 257 at blinded visits. The correlation between an open-label and a blinded visit is the AR(1) value times 0.976. The summed response is therefore AR(1) only within a phase. Under the two-constant form it is, in addition, AR(1) plus a white-noise term the analysis model omits. These statements follow from the algebra. Their effect on test calibration has not been measured separately.

Table 7 gives the feasibility bounds. At the starting $\rho = 0.8$ the two-constant form raises the bound only to 0.291. A single same-visit correlation repeated across visits that share little demands more of the biomarker row. In the open-label design, where every visit is on drug, it lowers the bound. The separable form raises the bound to 0.440. At $\rho = 0.7$ the two forms are close, 0.452 against 0.474. At the recommended ρ the separable form's gain in feasible range is therefore small, and its

Table 7: Largest feasible biomarker-response correlation c_{bm} under the decaying correlation, without carryover, minimum over the Hybrid, crossover and OL+BDC designs. Under compound symmetry the Hybrid and crossover designs tie for the minimum. Under the two AR(1) forms the minimum is attained in the Hybrid design at $\rho = 0.8$ and in the OL+BDC design at $\rho = 0.7$ (separable form: Hybrid 0.480), and the crossover design is never binding.

ρ_{cs} or ρ	Compound symmetry	AR(1), two-constant cross-component	Separable AR(1)
0.8	0.256	0.291	0.440
0.7	0.346	0.452	0.474

reasons are properties 1 and 4. The bound of 0.474 is attained in the OL+BDC design (0.480 in the Hybrid design). A correlation of 0.45 is feasible in every design without carryover. Effects as large as 0.6 exceed every feasibility bound in the Hybrid design.

The wider feasible range has a consequence that does not depend on the form. At the same c_{bm}, AR(1) data carry less information about the interaction than compound-symmetric data. The closed-form power of the paired-difference statistic in the Hybrid design at $c_{bm} = 0.25$ falls from 0.721 under the starting structure to 0.335 under the separable form at $\rho = 0.8$ and 0.290 at $\rho = 0.7$. A larger feasible correlation does not by itself mean more power. Effect sizes under different structures should therefore be compared on the scale of the implied slope, not of c_{bm}.

The recommended DGP uses the separable form with $c_1 = 0.2$. The choice rests on analyzability (property 4) and existence (properties 1 and 3). The value $c_1 = 0.2$ is the starting value and is uncalibrated.

Stage 6. Features the recommended DGP adds

The starting DGP correlates the biomarker with the drug response only. A biomarker that is also prognostic, through an association with the natural history, the placebo response or baseline severity, cannot be represented. A simulation built on the starting DGP therefore cannot show whether an analysis stays valid under such a biomarker. A prognostic effect of the biomarker is known to bias tests of a biomarker-by-treatment interaction when the analysis model omits or misrepresents it (49,50).

The recommended DGP adds optional correlations of the biomarker with the natural-history component (constant over visits) or the placebo-response component, a mean shift of the natural-history component that grows along its curve, and a correlation with baseline severity. It adds phase-specific

Table 8: Null rejection rates under a prognostic biomarker ($c_{bm} = 0$). Each pathway has strength 0.3: a correlation of 0.3 with the natural-history component (constant), the placebo-response component or baseline severity, and for the growing natural-history pathway a mean shift of the natural-history component of 0.3 of its SD (3 points) per SD of biomarker at the end of its curve, growing along the curve. 1,000 replicates per cell (two-MCSE band 0.036 to 0.064 for a single cell). Third run of study 3: the recommended DGP, with carryover decaying from the last on-drug mean and fixed path allocation. Rates above 0.075 in bold. The first run is in Table 17; the changes between runs are given in the text.

Analysis	Natural history, constant	Natural history, growing	Placebo response	Baseline severity
Binary-indicator analysis (random intercept, model-based)	**0.108**	0.056	**0.157**	0.064
Random intercept + CAR(1), CR2	**0.103**	0.049	**0.105**	0.052
The same, half-life chosen by AIC	**0.130**	0.059	**0.125**	0.057
Unstructured covariance, Kenward-Roger	**0.413**	**0.200**	**0.224**	0.048
The same, half-life chosen by AIC	**0.506**	**0.258**	**0.270**	0.054
Random intercept + CAR(1) with expectancy terms, CR2	0.045	0.040	0.052	0.055
The same, with phase-specific residual variances, CR2	0.056	0.037	0.051	0.059
The same, with phase-specific residual variances, model-based	0.054	0.042	0.049	0.059
Expectancy terms, half-life chosen by AIC, CR2	0.058	0.050	0.060	0.060
Random intercept + CAR(1) with free baseline slope, CR2	0.051	0.066	0.060	0.054

measurement error, a Weibull decay form and monotone dropout as further options.

A prognostic biomarker. We simulated a correlation of 0.3 between the biomarker and the natural-history component, the placebo-response component or baseline severity, and a growing mean shift of the natural-history component of 0.3 of its SD per SD of biomarker, with no interaction. Analyses that keep baseline as an outcome row use one biomarker main effect β_B for baseline and for the off-drug visits. When the biomarker is prognostic through a pathway that is absent at baseline, those two associations differ. Part of the difference is attributed to the interaction (Appendix A.4). Every analysis that shares the biomarker effect with baseline exceeded the nominal level in at least one such cell (Table 8).

The expectancy-term family stayed within the single-cell two-MCSE band in every prognostic cell. Its four analyses rejected 0.037 to 0.060. This family adds the expectancy and a biomarker-by-expectancy term to the mean model, with or without residual variances that differ by trial phase. The free-baseline model adds `bmc:bl0`, which gives baseline its own biomarker slope. It rejected

0.066 under the growing natural-history pathway, above the single-cell band, and 0.060 under the placebo pathway. Appendix A.4 explains why freeing the baseline slope alone is not sufficient when the prognostic slope grows over time or scales with the expectancy. A prognostic association with baseline severity alone did less harm, because baseline severity enters every visit equally. The binary-indicator analysis rejected 0.064 there. The scenarios in which the pathway is absent at baseline are the ones that show the inflation. Whether such a pathway, at a magnitude of 0.3, is realistic for blood pressure in PTSD is not known, but a DGP that cannot generate it cannot show the inflation.

Changes between runs. Study 3 was run three times (Methods). The reported run differs from the first only in the carryover rule, and from the second only in the allocation. Treating two runs as independent, the difference between them in one cell has a Monte Carlo SE of about 0.010 at a rate of 0.05. The first and third runs use the same seeds and allocation, so their differences are paired and that SE overstates their noise.

The carryover rule changed the prognostic table by at most 0.017, for the expectancy-term model with phase-specific variances (CR2) under the baseline-severity pathway (Appendix C, Table 17). Cells that crossed the bold threshold of 0.075: none. Cells that crossed the upper edge of the single-cell band: the free-baseline model under the growing natural-history pathway (first run 0.063, third run 0.066); the unstructured model with AIC-chosen half-life under the baseline-severity pathway (first run 0.064, third run 0.054).

Randomized allocation changed it by at most 0.041, for the unstructured model with AIC-chosen half-life under the placebo pathway. Cells that crossed the bold threshold: the binary-indicator analysis under the growing natural-history pathway (second run 0.077, third run 0.056). Cells that crossed the upper edge of the single-cell band: the binary-indicator analysis under the growing natural-history pathway (second run 0.077, third run 0.056); the CAR(1) model with AIC-chosen half-life under the growing natural-history pathway (second run 0.068, third run 0.059); the unstructured model with AIC-chosen half-life under the baseline-severity pathway (second run 0.065, third run 0.054). In all three runs the expectancy-term family stayed within the single-cell band in every prognostic cell, and the binary-indicator, CAR(1) and unstructured analyses, with or without an AIC-chosen half-life, exceeded it in at least one. Taken analysis by analysis, only the free-baseline model changed its classification. It stayed within the band in the first run and exceeded it under the growing pathway in the other two, by at most 0.009.

Feasibility with a prognostic correlation. A prognostic correlation occupies the same

Table 9: Largest feasible prognostic correlation given the biomarker-response correlation, and the reverse, under the recommended DGP (separable AR(1), $\rho = 0.7$, $c_1 = 0.2$), Hybrid design, minimum over its four paths. Correlation: the quantity whose bound is given. c_{PB}, c_{TV}, c_{BL}: correlation of the biomarker with the placebo-response component, the natural-history component and baseline severity. None: no value is feasible.

Correlation	Given	$t_{1/2} = 0.2$ week	$t_{1/2} = 0.5$ week
c_{PB} or c_{TV}	alone	0.513	0.513
	$c_{bm} = 0.25$	0.466	0.477
	$c_{bm} = 0.30$	0.435	0.451
	$c_{bm} = 0.45$	0.243	0.306
c_{BL}	alone	1.000	1.000
	$c_{bm} = 0.25$	0.857	0.876
	$c_{bm} = 0.30$	0.786	0.815
	$c_{bm} = 0.45$	0.374	0.496
c_{bm}	alone	0.485	0.518
	$c_{PB} = 0.3$	0.422	0.454
	$c_{PB} = 0.6$	none	none

biomarker row of the correlation matrix as c_{bm}. The two therefore share one feasibility condition. Writing the row as a sum of fixed patterns scaled by the correlations, the existence condition of Stage 1 becomes a quadratic form. The largest value of one correlation given the others is the positive root of a quadratic (Appendix A.4). Table 9 gives the bounds for the Hybrid design under the recommended DGP. At the stress test's half-life of 0.5 week, the prognostic correlation of 0.3 used above lies inside the feasible region at every c_{bm} up to 0.45, by 0.006 at 0.45. At 0.2 week it is feasible only up to $c_{bm} = 0.42$. A prognostic correlation of 0.6 is infeasible with any c_{bm}. That is the combination the companion study's two repaired runs had simulated (Stage 1). The correlations with the placebo-response and the natural-history components have identical bounds, because their patterns are the same constant on every visit of one component. A correlation with baseline severity has a looser bound, since baseline severity is uncorrelated with the response components and enters only through its square. The OL+BDC bounds are within 0.035 of the Hybrid values. The crossover bounds are looser on c_{bm} and c_{BL}, and slightly tighter on c_{PB} alone (0.487 against 0.513). We verified the bounds by checking that the smallest eigenvalue of the full matrix is positive at 0.999 times each bound and negative at 1.001 times it.

Size more generally. We pooled the 11 non-prognostic null cells (11,000 replicates). The nominal band is 0.05 plus or minus two Monte Carlo SE, $2\sqrt{0.05 \times 0.95/11{,}000} = 0.004$, that is 0.046 to 0.054. The binary-indicator analysis rejected 0.069, and 0.077 at $N = 35$. The unstructured model

with Kenward-Roger inference (51) rejected 0.055, with 0.084 at $N = 35$. Random-intercept models with continuous-time AR(1) residuals and CR2 cluster-robust standard errors (52,53), at the fixed analysis half-life, rejected 0.048 to 0.049. No single cell of these models exceeded 0.067. With the half-life chosen by AIC, the CAR(1) model rejected 0.053 and the expectancy-term model 0.056. The latter's largest single cell was 0.071, under $t_{1/2} = 1$ week. Of the analyses in this paragraph, the binary-indicator analysis, the unstructured model and the expectancy-term model with AIC-chosen half-life lie above the pooled band. Single-cell flags should be judged against the number of comparisons. With 10 reported analyses and 15 null cells, about 7 analysis-cell pairs would fall outside a two-MCSE band by chance at nominal size. The pairs are not independent, since all analyses share the same trials. A single exceedance, such as the free-baseline model under the growing pathway, is therefore weak evidence on its own.

The options. Table 10 shows what each optional feature did to size and power for three analyses: the binary-indicator analysis, the random-intercept CAR(1) model with CR2, and the expectancy-term model that the prognostic cells favor. Measurement error (SD 3 open label, 6 blinded) lowered power by 0.05 to 0.09. Dropout of 20% completely at random lowered power by 0.01 to 0.02. Dropout of 20% dependent on the last score lowered power by 0.04 to 0.06. A Weibull decay with the same half-life lowered power by 0.09 to 0.10 at shape 0.5 (a steep early decay) and raised power by 0.05 to 0.09 at shape 2. The true half-life itself moved power from 0.50 to 0.57 at zero to 0.35 to 0.42 at one week. Halving N gave power of 0.21 to 0.27, against 0.40 to 0.49 in the base cell. At $N = 150$ the CAR(1) model reached 0.79. In these null cells the CAR(1) and expectancy-term models rejected at most 0.067, against a single-cell band of 0.036 to 0.064 (the maximum, for the expectancy-term model under the base cell, lies 0.003 above the upper edge). The binary-indicator analysis rejected up to 0.078, under dropout completely at random. The features change power, which the planning section and the companion paper address.

Power and robustness. The expectancy-term model keeps its size under every prognostic pathway and has lower power. In the base cell its power was 0.404 against 0.486 for the CAR(1) model. A further 1,500 trials per cell under the second run's DGP (randomized allocation) were pooled with that run (2,000 per cell). There, size-adjusted power was 0.431 against 0.496, about 13% less (Additional file 4). Under compound symmetry at $c_{bm} = 0.25$ the order reverses. Power was 0.564 against 0.552, and in the pooled second-run trials size-adjusted power was 0.510 against 0.490. A DGP paper can say which analyses are valid under which features. The choice between power

Table 10: The optional features of the recommended DGP: null rejection rate ($c_{bm} = 0$; 1,000 replicates, MCSE 0.007) and power ($c_{bm} = 0.30$; 500 replicates, MCSE at most 0.022) for the binary-indicator analysis (Binary), the random-intercept CAR(1) model with CR2 and the expectancy-term model with CR2 (Expect.). Hybrid design, third run of study 3 (the recommended DGP: carryover decaying from the last on-drug mean, fixed allocation). Each row changes one feature of the base cell.

	Null rejection			Power		
Cell	Binary	CAR(1)	Expect.	Binary	CAR(1)	Expect.
Base cell ($N = 70$, $t_{1/2} = 0.5$, exponential, no dropout)	0.077	0.051	0.067	0.474	0.486	0.404
Measurement error (SD 3 and 6)	0.069	0.050	0.037	0.422	0.394	0.314
Dropout 20 percent, completely at random	0.078	0.053	0.055	0.466	0.464	0.392
Dropout 20 percent, dependent on last score	0.069	0.060	0.047	0.424	0.426	0.360
Weibull decay, shape 0.5	0.072	0.046	0.044	0.378	0.400	0.300
Weibull decay, shape 2	0.066	0.039	0.046	0.526	0.540	0.496
$t_{1/2} = 0$	0.071	0.040	0.041	0.536	0.566	0.498
$t_{1/2} = 1$ week	0.066	0.047	0.061	0.418	0.416	0.348
$N = 35$	0.077	0.053	0.048	0.270	0.246	0.206
$N = 150$	0.060	0.048	0.045	0.768	0.790	0.688

and robustness is the trialist's. It depends on whether a prognostic pathway absent at baseline is plausible for the biomarker in hand.

The recommended DGP includes the prognostic correlations as an option for that reason. The other three features are options whose effect on power is tabulated. The choice rests on analyzability. Measurement error is implemented in the stress-test driver and not yet in the package (Appendix B.5).

Stage 7. The recommended DGP

Table 11 collects the choices. Each row is marked as a change from the starting DGP, a choice among open alternatives, an option, or a retained element, and names the stage that justifies it. Values not listed are the starting values.

Table 11: The recommended data-generating process. The Default column gives the value of each element in the recommended DGP; where the stress test of Stage 6 used another value it is stated. The stress test's reported run uses fixed allocation; its second run assigned participants to paths at random, a driver option outside the recommended DGP (Appendix B.5).

Element	Specification	Default	Kind	Justified in
Components	$Y_{it} = \mathrm{BL}_i - (TV_{it} + PB_{it} + BR_{it})$, Gompertz means as in the starting DGP	starting parameters (Methods): BR $m = 10.99$, $r = 0.42$, SD 8; PB and TV $m = 6.51$, $r = 0.35$, SD $10\eta_t$ and 10; $d = 5$; biomarker 124.3 (SD 15.4); baseline 83.1 (SD 18.5)	retained	Methods
Feasibility	stop if the correlation matrix is not positive definite	always	change	Stage 1
Biomarker-response correlation	c_{bm} on drug; $c_{bm}2^{-t_{sd}/t_{1/2}}$ off drug after exposure; 0 before exposure	c_{bm} below the feasibility bound of every design in the study	change	Stage 2
Carryover on the mean	decay from the last on-drug value, $\mu_{BR,\mathrm{last}}\, 2^{-t_{sd}/t_{1/2}}$	$t_{1/2} = 0.1$ to 0.2 week (pharmacological, assumed); 0.5 in the stress test	choice	Stage 3

Element	Specification	Default	Kind	Justified in
Re-exposure	as in the starting DGP: time on drug restarts at zero	on, for titrated drugs	retained	Stage 3
Within-component correlation	AR(1) in weeks, $\rho^{\lvert w_t - w_s \rvert}$, one ρ for all three components	$\rho = 0.7$ (convention; report and vary)	choice	Stage 4
Cross-component correlation	separable, $c_1\rho^{\lvert w_t - w_s \rvert}$ at every lag; the Kronecker form applies to the correlation, since the PB SD scales with the expectancy	$c_1 = 0.2$ (starting value)	choice	Stage 5
Path allocation	fixed numbers, as in the starting DGP	starting values	retained	Methods
Prognostic correlations	correlation of the biomarker with TV or PB, or with baseline severity; or a mean shift of TV growing along its curve	off; 0.3 in the stress test; correlations must lie inside the joint bound with c_{bm} (Table 9)	option	Stage 6
Measurement error	independent, SD by phase	off; 3 open label, 6 blinded in the stress test	option (driver only)	Stage 6
Decay form	exponential or Weibull with the same half-life	exponential	option	Stage 6
Dropout	monotone, completely at random or at random given the last score	off; 20% in the stress test	option	Stage 6

Using the recommended DGP to plan a trial

What c_{bm} means. Conditional on the biomarker, the correlation encoding shifts the on-drug drug-response mean by $c_{bm}\sigma_{BR}$ per standard deviation of biomarker (Discussion). With the starting $\sigma_{BR} = 8$, $c_{bm} = 0.25$ is a slope of 2.0 symptom points per SD of biomarker, 0.35 is 2.8 points, 0.45 is 3.6 points, and the feasibility bound of 0.474 is 3.8 points. The biomarker's SD is 15.4 mm Hg in the starting parameters, so 0.25 is 0.13 points per mm Hg. The association first reported for prazosin is 14 points on the Clinician-Administered PTSD Scale per 10 mm Hg (12), or 21.5 points per SD of biomarker. Representing that slope under the correlation encoding would need $\sigma_{BR} \geq 45$ at the bound. That is larger than the SD of the symptom score itself (18.5 at baseline in the starting parameters). Under the starting parameters the bound is therefore below the reported association. The comparison is approximate for two reasons. The observed slope is on the total symptom change, not the drug-response component. And $\sigma_{BR} = 8$ is the starting choice, not an estimate.

That association did not replicate in our reanalysis of CSP 563 (Additional file 1). The interaction was +0.25 points per 10 mm Hg (95% CI −2.72 to 3.21). With $\sigma_{BR} = 8$ and the trial's biomarker SD of 14.2 mm Hg, this interval corresponds to c_{bm} from −0.57 to 0.48. The values of c_{bm} used in this paper, 0.25 to 0.45, imply 1.4 to 2.5 points per 10 mm Hg. They lie inside that interval. CSP 563 therefore neither supports nor excludes them, and it gives no reason to plan for a slope above the feasibility bound. A study that does need a larger slope should use the sensitivity model of the Discussion once it is implemented, or raise σ_{BR} with a stated reason.

What CSP 563 identifies. The reanalysis estimates some quantities of the DGP on the total symptom score, all as computed in Additional file 1 and not re-estimated here. The baseline SD is 16.0 (DGP: 18.5) and the biomarker SD 14.2 mm Hg (DGP: 15.4). The biomarker is not prognostic for baseline severity ($r = 0.02$, 95% CI −0.09 to 0.13), which supports a zero prognostic correlation with baseline. The placebo-arm improvement has the DGP's time course and is about 1.8 times its size. The SD of change from baseline grows from 17 to 24 points over 26 weeks, where the DGP holds it at 16.0. The total-score correlation decays by about 0.975 per week with about 10 percent measurement noise. The DGP's total-score correlation falls faster at short lags and then flattens. The trial does not identify the split between the natural-history and placebo-response components, the expectancy weight, the component-level correlations or the carryover half-life. All participants were blinded throughout, and no one stopped drug. A sensitivity arm of the DGP calibrated to

these estimates is future work. A working protocol for it exists in the repository, and it has not been run.

Sample size. Table 12 gives the total N for 80% power of the paired-difference statistic E9 under the recommended DGP, by design, c_{bm} and half-life, and the power at $N = 70$. The sample sizes are closed-form values for E9. In the Hybrid design E9 tracks the binary-indicator mixed model closely (Methods). Under compound symmetry it understates the binary-indicator mixed model in the crossover design. Under the recommended DGP the margin by which G10 exceeds E9 is smallest in the crossover design (below). At the pharmacological half-life the Hybrid design needs about 280 participants at $c_{bm} = 0.25$ and 84 at 0.45. At a 1-week half-life those rise to 480 and 146. For comparison, Brookes et al. (2) found that a main-effect test needs about a quarter of the sample of an interaction test of the same size. Their result is for a binary subgroup with two equal halves. Carrying the factor to a continuous biomarker slope of the same standardized size is an extrapolation.

The recommended analysis at the planned sample sizes. The companion paper (46) recommends the analysis G10 for the interaction. G10 is a linear mixed model with a random intercept and continuous-time AR(1) residual correlation. Its drug indicator is the exposure-decayed D_{bc} at an analysis half-life chosen by AIC over 0.25, 0.5, 1 and 2 weeks. The selection uses maximum-likelihood fits of the model without the interaction, so the choice of half-life does not depend on the term under test. The model at the chosen half-life is refitted by REML with the interaction, and the interaction is tested with CR2 cluster-robust standard errors and Satterthwaite degrees of freedom. The grid of half-lives does not include 0 or the pharmacological values of 0.1 to 0.2 week, so at those true values G10 uses an analysis half-life of at least 0.25 week. The AIC-chosen analyses of the stress test also consider 0 (Table 1). We simulated G10 under the recommended DGP with the starting parameters in every cell of Table 12, at $N = 70$ and at the E9 sample size for 80 percent power, with 1,000 replicates per cell. We also simulated the null ($c_{bm} = 0$) at $N = 70$ for each design and half-life with 2,000 replicates. The last column of Table 12 gives G10 power at $N = 70$ at the 0.2-week half-life, and Table 13 gives every cell. Where the E9 sample size exceeded 400, or was above 800 and not computed, the cell ran at $N = 400$ and is marked.

Three results follow. First, G10 rejected the null at 0.046 to 0.053 across the nine design-by-half-life cells (Monte Carlo SE at most 0.005). Second, at $N = 70$ G10 has power more than two Monte Carlo SEs above that of E9 in 23 of the 27 cells, within two SEs in 4 and below in none. In the

Hybrid design at $c_{bm} = 0.25$ and the 0.2-week half-life, G10 power is 0.357 (MCSE 0.015) against 0.285 for E9. In the crossover design at the same cell it is 0.243 (MCSE 0.014) against 0.202. Third, at the E9 sample size for 80 percent power, G10 power is 0.844 to 0.936 in the 20 uncapped cells, and 0.601 to 0.915 in the 7 cells that ran at $N = 400$. Under the recommended DGP at $\rho = 0.7$ with the starting parameters, the closed-form E9 rule is therefore conservative for G10. A trial sized by Table 12 and analyzed with G10 had power of at least 0.844 in every uncapped cell, with the smallest margin in the crossover design. Fits failed to converge in at most 0.008 of replicates in any cell, and the rejection rates are computed over converged fits.

The same simulated trials were also analyzed with G9, which chooses the half-life by AIC with the interaction in the model. In every cell the G9 and G10 rejection rates differ by at most 0.040.

The sample size for G10. In the nine Hybrid cells we simulated G10 at 6 multiples of the E9 sample size, from 0.35 to 1.25 times, with 500 replicates at each, and fitted a probit regression of rejection on $\log N$ in each cell. Table 14 gives the N at which the fitted power is 0.80. That N lies inside the simulated range in all 9 cells, and it is 0.71 to 0.83 times the E9 value. The E9 values are closed form, so this comparison is the measured difference between the paired-difference approximation and the analysis a trial would use.

Serial correlation. The planning results above assume $\rho = 0.7$, a convention (Stage 4). We repeated the closed-form calculation and the G10 simulation at $\rho = 0.5$ and 0.9, with the same seeds and replicate counts. At 0.5 the grid of c_{bm} is unchanged, and one cell lies above its bound (OL+BDC at 0.45 without carryover). At 0.9 the Hybrid bound falls to 0.340, so the grid moves to 0.20, 0.25 and 0.30. Table 15 gives the cells at the 0.2-week half-life.

In the Hybrid design the required sample size depends strongly on ρ above 0.7 and little below it. At $c_{bm} = 0.25$ the E9 rule gives 314, 278 and 154 participants at $\rho = 0.5$, 0.7 and 0.9. The crossover needs 1.5 times the Hybrid sample at 0.7 and 2.6 times at 0.9. The likely reason is visit spacing. The Hybrid design compares on- and off-drug visits one week apart, where a high serial correlation cancels most of the shared noise. The crossover compares two blocks of four visits 2.5 weeks apart, ten weeks apart on average. G10 rejected the null at 0.047 to 0.056 in the 18 design-by-half-life cells at the two new values. At the E9 sample size its power was 0.815 to 0.867 at $\rho = 0.5$ and 0.938 to 0.999 at 0.9, in the uncapped cells. The E9 rule therefore stays conservative for G10, and at $\rho = 0.9$ it is very conservative. There G10 had power 0.751 in the Hybrid design at $c_{bm} = 0.25$ with only 70 participants. The G10 sample-size search, repeated in the Hybrid design, gives 0.80 to

0.91 times the E9 value at $\rho = 0.5$ and 0.45 to 0.57 times at 0.9, against 0.71 to 0.83 at 0.7 (Table 14; cells whose value lies outside the simulated range are left out of these ranges: 0 at the two new values).

In the crossover design at $\rho = 0.9$, 8 to 12 percent of G10 fits failed in each cell at $N = 70$, and 13 to 19 percent at the E9 sample sizes below 400. In the cells that ran at $N = 400$ at most 0.1 percent failed. In every other cell at the two new values at most 2.6 percent failed. In the failed replicates we refitted, the model chosen by AIC put the serial correlation at its upper limit of 1 and the random-intercept SD near zero. A correlation that does not decay cannot be told apart from a random intercept, and the CR2 step then failed because the fitted covariance was singular. The rates in Table 15 are over the fits that succeeded. A trialist who expects slow decay and plans a crossover analyzed with G10 should expect such failures and specify a fallback model in advance.

When these sample sizes apply. The G10 sample sizes assume that the biomarker has no prognostic pathway. We fitted G10 to the trials of the stress test (Stage 6). It rejected the null at 0.052 in the base cell, and at 0.108, 0.051, 0.110 and 0.050 under the constant natural-history, growing natural-history, placebo and baseline-severity pathways (single-cell band 0.036 to 0.064). The expectancy-term model rejected 0.040 to 0.055 in the same cells (Table 8). G10 therefore holds its level when the biomarker acts through baseline severity or a growing natural history, but not when it acts through a constant natural history or the placebo response. A trial in which either pathway is plausible should be planned for the expectancy-term analysis. On the power-curve trials of the stress test, matching G10's power with that analysis needs about 1.13 to 1.27 times the G10 sample size, by the normal approximation to power, at c_{bm} from 0.225 to 0.45. At $c_{bm} = 0.15$ both analyses have power near 0.15 and the factor, 0.96, is within Monte Carlo error of 1. The G10 sample sizes of Tables 13 and 14 can be scaled by that factor. The factor comes from one design, half-life and ρ, the stress test's base cell.

Parameters by status.

- *Starting values:* the Gompertz parameters of the three components and their standard deviations; the biomarker and baseline means and standard deviations; $c_1 = 0.2$; the path allocation; the expectancy values; the restart of time on drug at re-exposure.
- *Chosen here:* the decaying correlation and the stop on infeasible matrices; carryover decaying from the last on-drug mean; AR(1) in weeks with $\rho = 0.7$; the separable cross-component

Table 12: Planning under the recommended DGP (separable AR(1), $\rho = 0.7$, $c_1 = 0.2$, decaying correlation, carryover decaying from the last on-drug mean, starting response parameters): total N for 80 percent power of the paired-difference statistic E9 at $\alpha = 0.05$, by design, interaction strength and carryover half-life, with the implied slope in symptom points per SD of biomarker ($\sigma_{BR} = 8$) and the power at $N = 70$ and the 0.2-week half-life for E9 (closed form) and for the analysis G10 (simulated, 1,000 replicates per cell, Monte Carlo SE at most 0.016). Participants are allocated to paths in the original fixed proportions.

			N for 80 percent power			Power at $N = 70$, 0.2 week	
Design	c_{bm}	Slope (points per SD)	$t_{1/2} = 0$	0.2 week	1 week	E9	G10
Hybrid	0.25	2.0	272	278	480	0.285	0.357
	0.35	2.8	138	140	244	0.503	0.610
	0.45	3.6	82	84	146	0.728	0.836
OL+BDC	0.25	2.0	394	404	> 800	0.210	0.275
	0.35	2.8	200	206	452	0.368	0.507
	0.45	3.6	120	124	272	0.558	0.722
Crossover	0.25	2.0	424	424	454	0.202	0.243
	0.35	2.8	216	216	230	0.353	0.365
	0.45	3.6	130	130	138	0.537	0.592

Table 13: Rejection rate of the analysis G10 under the recommended DGP of the preceding planning table, by design, interaction strength and carryover half-life, at $N = 70$ and at the total N the closed-form E9 rule gives for 80 percent power (the preceding table). Rows with $c_{bm} = 0$ give the size at $N = 70$ (2,000 replicates per cell, Monte Carlo SE at most 0.005); the other rows give power (1,000 replicates per cell, Monte Carlo SE at most 0.016). †: the E9 sample size exceeds 400 and the cell ran at $N = 400$. Rejection rates are over converged fits (at most 0.8 percent of replicates failed in any cell).

		At $N = 70$			At the E9 N for 80 percent power		
Design	c_{bm}	$t_{1/2} = 0$	0.2 week	1 week	$t_{1/2} = 0$	0.2 week	1 week
Hybrid	0.00	0.046	0.051	0.048			
	0.25	0.367	0.357	0.217	0.915	0.900	0.809†
	0.35	0.616	0.610	0.362	0.908	0.899	0.885
	0.45	0.855	0.836	0.531	0.894	0.903	0.874
OL+BDC	0.00	0.052	0.051	0.051			
	0.25	0.295	0.275	0.133	0.936	0.915†	0.601†
	0.35	0.495	0.507	0.235	0.934	0.936	0.866†
	0.45	0.725	0.722	0.362	0.930	0.932	0.891
Crossover	0.00	0.053	0.046	0.050			
	0.25	0.187	0.243	0.220	0.842†	0.840†	0.812†
	0.35	0.356	0.365	0.378	0.861	0.867	0.862
	0.45	0.594	0.592	0.551	0.860	0.869	0.844

Table 14: Total N for 80 percent power in the Hybrid design under the recommended DGP with the starting parameters: closed-form E9 (Table 12; at $\rho = 0.5$ and 0.9, Table 15) and the analysis G10. G10 was simulated at 6 values of N per cell at $\rho = 0.7$ (500 replicates each), at the same multiples of the E9 value at 0.5, and at lower multiples at 0.9, where G10 needs far fewer participants than E9. The N at which a probit regression of rejection on $\log N$ reaches 0.80 is rounded up.

ρ	c_{bm}	$t_{1/2}$ (weeks)	E9 (closed form)	G10 (simulated)	G10 / E9	Simulated N
0.7	0.25	0	272	193	0.71	95 to 340
	0.25	0.2	278	205	0.73	97 to 348
	0.25	1	480	384	0.80	168 to 600
	0.35	0	138	104	0.75	48 to 172
	0.35	0.2	140	108	0.77	49 to 175
	0.35	1	244	193	0.79	85 to 305
	0.45	0	82	61	0.74	29 to 102
	0.45	0.2	84	64	0.76	29 to 105
	0.45	1	146	121	0.83	51 to 182
0.5	0.25	0	308	271	0.88	108 to 385
	0.25	0.2	314	268	0.85	110 to 392
	0.25	1	542	434	0.80	190 to 678
	0.35	0	156	142	0.91	55 to 195
	0.35	0.2	158	140	0.88	55 to 198
	0.35	1	276	244	0.88	97 to 345
	0.45	0	94	83	0.88	33 to 118
	0.45	0.2	96	86	0.89	34 to 120
	0.45	1	166	146	0.88	58 to 208
0.9	0.20	0	238	110	0.46	36 to 238
	0.20	0.2	242	117	0.48	36 to 242
	0.20	1	420	240	0.57	63 to 420
	0.25	0	152	68	0.45	23 to 152
	0.25	0.2	154	74	0.48	23 to 154
	0.25	1	268	153	0.57	40 to 268
	0.30	0	104	50	0.47	20 to 104
	0.30	0.2	106	53	0.50	20 to 106
	0.30	1	186	107	0.57	28 to 186

Table 15: Planning by serial correlation at the 0.2-week half-life under the recommended DGP (separable AR(1), $c_1 = 0.2$, starting response parameters): the closed-form total N for 80 percent power of E9, and G10 power at $N = 70$ and at that N, with the percent of G10 fits that failed at $N = 70$. 1,000 replicates per cell. The rows at $\rho = 0.7$ repeat Table 12. At $\rho = 0.9$ the grid of c_{bm} lies inside the lower bound. Not feasible: c_{bm} lies above the bound of that design. †: the E9 sample size exceeds 400 and the cell ran at $N = 400$. Power is over the fits that succeeded.

				G10 power		
ρ	Design	c_{bm}	E9 N	At $N = 70$	At the E9 N	Failed (%)
0.5	Hybrid	0.25	314	0.258	0.849	0
		0.35	158	0.487	0.829	0
		0.45	96	0.693	0.822	0
	OL+BDC	0.25	412	0.225	0.844†	0
		0.35	208	0.391	0.859	0
		0.45	126	0.606	0.856	0
	Crossover	0.25	348	0.267	0.864	0
		0.35	176	0.405	0.825	0
		0.45	106	0.623	0.815	0
0.7	Hybrid	0.25	278	0.357	0.900	0
		0.35	140	0.610	0.899	0
		0.45	84	0.836	0.903	0
	OL+BDC	0.25	404	0.275	0.915†	0
		0.35	206	0.507	0.936	0
		0.45	124	0.722	0.932	0
	Crossover	0.25	424	0.243	0.840†	0
		0.35	216	0.365	0.867	0
		0.45	130	0.592	0.869	0
0.9	Hybrid	0.20	242	0.546	0.979	0
		0.25	154	0.751	0.976	0
		0.30	106	0.897	0.979	0
	OL+BDC	0.20	462	0.416	0.992†	0
		0.25	296	0.627	0.999	0
		0.30	204	0.757	0.997	0
	Crossover	0.20	628	0.212	0.803†	10
		0.25	402	0.315	0.944†	8
		0.30	278	0.451	0.938	9

form; the symptom half-life range of 0.1 to 0.2 week; the option defaults used in the stress test.

- *Uncalibrated:* every value above. None was re-estimated for this paper. CSP 563 estimates the baseline and biomarker SDs, the total placebo-arm improvement and the total-score covariance, and it does not identify the component split, the expectancy weight or the half-life (above). The values to which the results are most sensitive are ρ (Stage 4), σ_{BR} (which sets the slope a given c_{bm} implies) and the half-life.

Choosing c_{bm}. Take the smallest feasibility bound over the designs and half-lives in the study (Table 7 and Appendix A.5; 0.474 for the three designs at $\rho = 0.7$ without carryover, higher with carryover). State c_{bm} as a slope in outcome points per SD of biomarker. Keep it below that bound. The package reports an infeasible value by stopping.

Discussion

The correlation encoding of the interaction

Conditional structure. Under joint normality, conditioning on the biomarker shows what the correlation encoding does to the drug response. With b_i the standardized biomarker,

$$E(BR_{it} \mid B_i) = \mu_{BR,t} + c_{bm}\, g_t\, \sigma_{BR}\, b_i, \qquad \mathrm{Cov}(BR_{i\cdot} \mid B_i) = \Sigma_{BR} - c_{bm}^2 \sigma_{BR}^2\, g g^\top.$$

The encoding therefore shifts the conditional drug-response mean in proportion to the correlation profile g_t. It reduces the residual drug-response covariance by a rank-one term. Two consequences follow.

- **The power loss under carryover is set by the profile.** The size of the loss depends on whether the interaction fades with the drug effect, which is the correlation rule.
- **The feasibility bound is the condition** that the reduced residual covariance remains positive definite.

The correlation encoding is therefore a parameterization of the interaction as a correlation. It keeps the marginal variance of the drug response fixed. Outside the starting DGP we found no use of it in the N-of-1 literature we searched (Background). Its strengths and its limits both follow from it.

The correlation encoding has four strengths.

- **One interpretable parameter.** The interaction strength is a single correlation. Its square, c_{bm}^2, is the share of on-drug drug-response variance the biomarker explains, the systematic part of treatment-effect variation in the sense of Ding et al. (54).
- **A reduced form of responder heterogeneity.** Suppose participants fall into two latent responder classes, and both the biomarker and the drug response differ between classes. Then class membership carries the covariance between them. Matching second moments gives $c_{bm}^2 = f_B f_{BR}$, where f_B and f_{BR} are the fractions of the biomarker's and the drug response's variance that lie between classes (Appendix A.6). The encoding therefore matches the second moments of a responder-class mechanism without requiring the mixture to be identified. Identifying the mixture typically needs larger samples than N-of-1 programs enroll (55–57). The match is limited to second moments. The mixture's conditional mean is nonlinear in the biomarker, and its higher moments differ.
- **The marginal variance of the drug response is fixed.** The biomarker explains part of the existing variation rather than adding to it. The effect size can therefore be stated as a share of response variance without changing the response distribution.
- **It fits a biomarker that marks individual sensitivity to the drug** better than one that sets a fixed dose-response shift.

It also has four structural limits.

- **The joint distribution must exist,** which bounds the effect size. In outcome units the bound is $c_{bm}^* \sigma_{BR}$ per SD of biomarker, 3.8 points under the recommended DGP at the starting σ_{BR}. The planning section shows this is below the association first reported for prazosin, which did not replicate in CSP 563. The bound depends on the correlation structure and the design.
- **The correlation must be specified at off-drug visits.** A constant correlation after first exposure removes the interaction under carryover. The specification adopted here is itself asymmetric between titration and washout (Stage 2).
- **The drug-response component keeps its full variance at off-drug visits,** with zero mean. This adds noise to every on-off contrast. It belongs to the component decomposition, not to the correlation encoding.

- **A correlation is less intuitive than a slope for clinical readers.** The slope it implies, $-c_{bm}\sigma_{BR}/\sigma_{bm}$, must be derived before it can be compared with an observed effect. The comparison depends on σ_{BR}, which is not estimated.

A model that generates the drug response from a participant-level sensitivity to exposure would address these limits:

$$BR_{it} = \omega_i f(t) + \varepsilon_{it}, \qquad \omega_i = \bar{\omega}\,(1 + \zeta b_i) + u_i,$$

where $f(t)$ is the exposure-response time course, ζ the biomarker's effect on sensitivity and u_i the sensitivity it does not explain. This is a random-coefficient model with a covariate-by-exposure interaction, the form used to describe individual response in replicated crossover and N-of-1 trials (7,28).

- It encodes the biomarker through individual sensitivity, so it has no feasibility bound.
- Its moderation is proportional to the response at every visit, so it fades during titration as well as after discontinuation.
- Its off-drug variance shrinks with the response.

It is not yet implemented in the software. Its response half-life for prazosin in PTSD is not known. Until then, the correlation encoding's range of use is effect sizes below the feasibility bound, which the planning section translates into outcome units.

We keep the correlation encoding in the recommended DGP because it is the starting point. Each change can then be traced stage by stage. We do not claim that the correlation encoding is preferable to the random-coefficient form. Whether the two give the same power, feasibility and design ranking at a matched implied slope has not been tested.

What remains uncalibrated

The recommended DGP meets the existence and identifiability criteria by construction and derivation. It meets the analyzability criterion in part: the separable form matches the analysis model within each phase, and the effect of the departure across phases on test calibration has not been measured. It meets the fidelity criterion by judgment. It does not meet the calibration criterion.

Every parameter is the starting value or a choice made here. The values that matter most are the serial correlation, the drug-response SD and the symptom half-life. The parallel-group trial CSP 563 informs only the first, and only on the total score. Its CAPS correlation decays slowly, and the DGP's total-score correlation decays faster at short lags (planning section). In the reanalysis (Additional file 1), compound symmetry fit it worst among the structures compared. All three values could be estimated from an aggregated N-of-1 trial of prazosin with weekly symptom measurement. Such a trial would also show which correlation form fits weekly series.

Recommendations

For simulation studies of these designs, we recommend the following.

1. **Let the biomarker-response correlation decay after discontinuation,** and stop on matrices that are not positive definite.
2. **State the choices:** carryover decaying from the last on-drug mean, AR(1) in weeks at a reported ρ, and the separable cross-component form. Repeat the analysis at neighboring ρ where the conclusion could depend on it.
3. **Choose c_{bm} below the feasibility bound** for every design in the study. Report it on two scales: as c_{bm}, and as the implied slope in outcome points per SD of biomarker.
4. **Keep the starting DGP as a reference arm** when results are compared with earlier ones. Attribute each difference to a named stage.
5. **Include a prognostic scenario** whenever the analysis shares the biomarker effect with the baseline visit. State the trade-off between the power of the plain model and the robustness of the expectancy-term model rather than resolving it for the trialist.
6. **Use the sensitivity model, once available, for effects above the feasibility bound.** Until then, state the bound in outcome units as a limitation.

Limitations

- **The closed form covers one statistic.** It is exact for the paired-difference statistic, not for the binary-indicator mixed model. The two agree closely in the Hybrid design. Under compound symmetry, in the crossover design, the closed form understates the binary-indicator mixed model. The sample sizes of the planning table are E9 values. The simulated G10

columns show the direction and size of the difference under the recommended DGP: at the E9 sample size G10 power was 0.844 to 0.936 in the uncapped cells. The G10 sample size for 80 percent power was searched for in the Hybrid design only, by simulation at 6 values of N per cell, and it carries Monte Carlo error that Table 14 does not show.

- **The planning results depend on ρ and assume no prognostic pathway.** The main tables use $\rho = 0.7$. The one available PTSD series suggests a slower decay (Stage 4). At $\rho = 0.9$ the Hybrid design needs about half the sample, the largest feasible effect is lower, and G10 fits fail often in the crossover design (Table 15). The G10 sample sizes assume no prognostic pathway; the expectancy-term analysis that keeps its size under one needs a larger sample (planning section). The factor is estimated in one cell of the stress test only.
- **At pharmacological half-lives the recommended DGP shows no carryover effect.** The on/off correlation difference is within 0.01 of 1 at 0.2 week. The DGP therefore tests carryover only at half-lives of 0.5 week and above, which we cannot ground pharmacologically.
- **The correlation is asymmetric between titration and washout.** Full moderation during the first weeks on drug is a modeling choice inherited from the starting DGP, not a consequence of the pharmacology.
- **Parameter values are not estimated.** The Gompertz parameters, correlations, σ_{BR} and prognostic correlations are starting values or chosen values. A reanalysis of CSP 563 estimates the total-score variance and correlation, which differ from the DGP's, and cannot identify the component split, the expectancy weight or the half-life. The motivating blood-pressure association did not replicate there.
- **The DGP is Gaussian with a single biomarker.**
- **The pharmacological carryover range** for prazosin at weekly visits is inferred from its plasma half-life, not measured as a symptom-response half-life.
- **The stress test is exploratory.** Cells and analyses were added after first results. Its prognostic scenarios are the ones that show the size inflation by construction.
- **The literature search is incomplete.** We searched PubMed and Europe PMC, not Web of Science or Scopus, and the searches missed one relevant study that we found by other means (Background). A prior use of the correlation encoding may therefore exist.
- **The correlation encoding is kept from the starting DGP.** The random-coefficient form of the Discussion has no feasibility bound and no off-drug drug-response noise. We have not compared the two at a matched implied slope.

Conclusions

We set out to move toward an optimal DGP for simulating predictive-biomarker interactions in aggregated N-of-1 trials. We took its elements one at a time, and three choices matter most. First, the biomarker-response correlation must decay after discontinuation. If it stays constant after first exposure, the interaction is absent among the post-baseline visits under any carryover. Power then falls nearly to the test size. With a decaying correlation, power returns to its no-carryover value at pharmacological half-lives (the on/off correlation difference is within 0.01 of 1 at 0.2 week). It declines smoothly at half-lives of 0.5 week and above. The decline is proved monotone for the path-stratified paired-difference statistic and was checked on a grid for the pooled statistic. Second, the serial correlation structure sets the largest interaction the DGP can hold. A separable AR(1) structure at a serial correlation of 0.7 allows a biomarker-response correlation of 0.47. That is 3.8 symptom points per SD of biomarker at the starting drug-response SD. The value 0.7 is a convention, however, and the one available PTSD series suggests a slower decay. At 0.9 the bound falls to 0.340, and the Hybrid design needs 154 participants at $c_{bm} = 0.25$ instead of 278. Third, a matrix outside the bound should be refused rather than repaired.

The planning table gives the power at matched strength. It does so for the paired-difference statistic in closed form and for the recommended mixed-model analysis G10 by simulation. At the closed-form sample size, G10 power was 0.844 to 0.936 in the uncapped cells. In the Hybrid design the G10 sample size for 80 percent power was 0.71 to 0.83 times the closed-form value. We propose the recommended DGP as a reference for simulation studies of these designs. It has options for prognostic pathways and other features that change which analysis is valid. Every parameter is labeled as a starting value, a choice or uncalibrated. Calibrating the serial correlation, the drug-response SD and the symptom half-life is the next step toward an optimal DGP.

Glossary

Terms are listed in the order in which the paper introduces them. Labels in backticks are the identifiers used in the repository's code, data files and table headings.

- **Starting DGP** (`published` in code and data labels): the data-generating process of Hendrickson et al. (2020), as implemented in their code at commit 58b32a9.

- **Recommended DGP** (`refined` in code and data labels): the data-generating process of Table 11, which adds the decaying correlation, the feasibility check, carryover decaying from the last on-drug mean, the separable AR(1) structure and the optional features to the starting DGP.
- **Correlation encoding of the interaction** (`mvn`): the predictive biomarker is a correlation between the biomarker and the drug response that is present on drug and absent off drug.
- **Biomarker-response correlation** (c_{bm}, `c.bm`): the correlation between the baseline biomarker and the drug-response component at an on-drug visit, which sets the strength of the interaction.
- **Correlation profile** (g_t): the fraction of c_{bm} present at visit t, equal to 1 on drug.
- **Constant rule** (`step` in code and data labels): the biomarker-response correlation is c_{bm} at every visit where the drug-response mean is nonzero, and so is constant after first exposure.
- **Decaying correlation** (`graded` in code and data labels): the decaying rule, under which the correlation is c_{bm} on drug, $c_{bm}2^{-t_{sd}/t_{1/2}}$ off drug after exposure, and 0 before exposure.
- **On/off correlation difference** ($\mathcal{G}$): the on-drug value of g_t minus the mean of g_t over the post-baseline off-drug visits, $1 - \bar{g}_{\text{off}}$, through which alone the correlation rule affects closed-form power.
- **Feasibility bound** (c^*_{bm}, "Ceiling" in tables): the largest c_{bm} for which the correlation matrix is positive definite under a given correlation structure, design and correlation profile.
- **Carryover half-life** ($t_{1/2}$, `carryover_t1half`): the time after discontinuation for the drug-response mean, and under the decaying rule the biomarker-response correlation, to fall by half.
- **Pharmacological and behavioral half-lives**: half-lives of 0.1 to 0.2 week are the assumed pharmacological range for prazosin, and half-lives of 0.5 week and above are labeled behavioral scenarios.
- **Compounding carryover rule** (`recursive`): the drug-response mean at an off-drug visit is the previous visit's mean decayed by the time since discontinuation, so the decay compounds across visits.
- **Anchored carryover rule** (`anchored`): the drug-response mean at an off-drug visit is the last on-drug mean decayed by the elapsed time since discontinuation.
- **Drug-response component** (BR): the latent response driven by time on drug, with mean on a modified Gompertz curve.

- **Placebo-response component** (PB): the latent response driven by time in the trial and scaled by the expectancy.
- **Natural-history component** (TV): the latent response driven by time in the trial that does not depend on treatment.
- **Baseline severity** (BL): the participant's symptom level before any response, which enters every visit equally.
- **Expectancy** (η_t): the participant's probability of being on drug at visit t given the design, 1 in the open-label phase and 0.5 when blinded.
- **Compound symmetry** (ρ_{cs}): the same correlation between any two visits of a component, whatever their separation.
- **Separable AR(1) structure**: the same first-order autoregressive correlation $\rho^{|w_t - w_s|}$ for every component, with a constant between-component correlation c_1 at every lag, so that the response block is a Kronecker product.
- **Two-constant form**: an AR(1) structure in which the same-visit cross-component correlation c_1 and the other-visit correlation $c_\times \rho^{|w_t - w_s|}$ use different constants.
- **Hybrid design** (`Hybrid`): open-label titration, blinded discontinuation, then a brief crossover, with visits at weeks 4, 8, 9, 10, 11, 12, 16 and 20.
- **OL+BDC** (`OLBDC`): open label followed by blinded discontinuation.
- **CO**: the crossover design, in which participants receive drug and placebo in one of two orders.
- **OL**: the open-label design, in which every visit is on drug.
- **Path**: one sequence of drug and placebo phases within a design, to which participants are allocated in fixed numbers.
- **Paired-difference statistic** (E9, E9b): the per-participant difference between the mean on-drug and mean off-drug scores, regressed on the biomarker, using post-baseline visits only (E9) or counting baseline as one more off-drug visit (E9b).
- **Binary-indicator analysis**: the linear mixed model used with the starting DGP, with a random intercept, a binary drug indicator and baseline included as an off-drug row.
- **G10** (`E7null_cr2` in data labels): the analysis the companion paper recommends. It is a random-intercept CAR(1) mixed model with the exposure-decayed drug indicator. The half-life is chosen by AIC over 0.25, 0.5, 1 and 2 weeks from maximum-likelihood fits without the interaction. The model is then refitted by REML with the interaction and tested with CR2 cluster-robust standard errors and Satterthwaite degrees of freedom.

- **G9** (`E7cr2` in data labels): as G10, except that the half-life is chosen with the interaction in the model.
- **Capped cell** († in Table 13): a cell whose E9 sample size for 80 percent power exceeds 400, simulated at $N = 400$.
- **Exposure-decayed drug indicator** (D_{bc}): a drug indicator equal to 1 on drug, $2^{-t_{sd}/t_{1/2}}$ off drug after exposure, and 0 before exposure.
- **Prognostic correlation** (c_{PB}, c_{TV}, c_{BL}; `prog-PB`, `prog-TV-const`, `prog-TV-grow`, `prog-BL` in data labels): a correlation of the biomarker with the placebo-response component, with the natural-history component (constant over visits), or with baseline severity. The growing natural-history pathway (`prog-TV-grow`) is not a correlation but a mean shift of the natural-history component that grows along its curve.
- **Expectancy-term model**: the random-intercept CAR(1) analysis that adds the expectancy and a biomarker-by-expectancy term to the mean model.
- **Free-baseline model**: the random-intercept CAR(1) analysis that adds `bmc:bl0`, a separate biomarker slope for the baseline row.
- **CAR(1)**: the continuous-time first-order autoregressive residual correlation `corCAR1` of the analysis model.
- **Stress test** (`11-strawman-stress-test.R`): study 3, the simulation that fits fourteen analyses to trials from the recommended DGP with and without a prognostic biomarker. The ten analyses of Table 8 are reported.
- **Runs of the stress test** (`summary-full*.csv`, `summary-full*-anchored.csv` and `summary-full*-anchored-fixed-ha0.5.csv`): the first run used the compounding carryover rule (`recursive`) and fixed allocation. The second used carryover decaying from the last on-drug mean (`anchored`) and randomized allocation. The third, reported in Stage 6, used the anchored carryover rule and fixed allocation, the recommended DGP as defined.
- **The five criteria**: existence (the joint distribution exists), identifiability under carryover (the interaction remains an on-minus-off contrast), fidelity to the pharmacology, analyzability (the analysis model can represent the generated structure and the DGP can generate the features that decide validity), and calibration status (each parameter is labeled as a starting value, a choice or uncalibrated).

Abbreviations

ADEMP: aims, data-generating mechanisms, estimands, methods, performance measures; AIC: Akaike information criterion; AR(1): first-order autoregressive; BDC: blinded discontinuation; BL: baseline severity; BR: drug-response component; CAPS: Clinician-Administered PTSD Scale; CAR(1): continuous-time first-order autoregressive correlation; CI: confidence interval; CO: crossover; CR2: bias-reduced cluster-robust variance estimator; CS: compound symmetry; CSP: Cooperative Studies Program; DGP: data-generating process; E9, E9b: the paired-difference interaction statistic on post-baseline visits, and with baseline counted as an off-drug visit; MCSE: Monte Carlo standard error; ML: maximum likelihood; OL: open label; OL+BDC: open label followed by blinded discontinuation; PB: placebo-response component; PTSD: post-traumatic stress disorder; REML: restricted maximum likelihood; SD: standard deviation; SE: standard error; TV: natural-history component; VA: Department of Veterans Affairs.

Supplementary material

The additional files are in the `supplement/` directory of this manuscript in the repository. They are project documents, not peer-reviewed publications. Each is cited for the result named here.

- **Additional file 1** (`S1-csp563-reanalysis.pdf`). The reanalysis of CSP 563: the arm-by-blood-pressure interaction, the biomarker's association with baseline severity, and the variance and serial correlation of the total symptom score. Aggregate results only; the participant-level data are not public.
- **Additional file 2** (`S2-stress-test-protocol.pdf`). The working protocol of the stress test (study 3).
- **Additional file 3** (`S3-analysis-selection.pdf`). The post hoc comparison and selection of analyses from the stress test.
- **Additional file 4** (`S4-stress-test-results.pdf`). Stress-test results, including the size-adjusted power comparison of the expectancy-term and CAR(1) models.
- **Additional file 5** (`S5-compendium-mathematics.pdf`). The compendium of mathematical results, including the derivations behind Appendix A.4.
- **Additional file 6** (`S6-separable-covariance.pdf`). The full derivation and numerical check of Appendix A.5.

- **Additional file 7** (`S7-carryover-recursion.pdf`). The carryover recursion in the starting code (Appendix B.3).

Declarations

Ethics approval and consent to participate. Not applicable: the study uses simulated data only.

Consent for publication. Not applicable.

Use of CSP 563 results. Additional file 1 reports aggregate results from a reanalysis of VA Cooperative Studies Program trial CSP 563. No participant-level data are shared. [**Placeholder: confirmation that publication of these aggregate results is permitted under the data use agreement is pending.**]

Availability of data and materials. All code and simulated data are in the pmsimstats-ng repository (`https://github.com/rgt47/pmsimstats-ng`). A tagged release with a persistent identifier will be deposited at submission. Until then, cite commit eafe30762, from which this version was rendered with uncommitted changes; a submitted version must be rendered from a clean commit.

- the closed-form calculations, in `analysis/scripts/quick-sim/carryover-closed-form/01-e9-closed-fo`
  and the serial-correlation stage, planning table and trajectory figure, in `17-rho-stage.R`
  in the same directory, and the joint feasibility bounds with a prognostic correlation in
  `18-prognostic-joint-ceiling.R`;
- the G10 and G9 simulations at the planned sample sizes, on the Figure 4B grid and for the
  G10 sample size, in `19-g9-planning-and-fig4b.R` in the same directory (run with `--params`
  `published --specs E7null_cr2,E7cr2`, blocks P and F in one job and block N in another),
  with their outputs under `analysis/data/quick-sim/carryover-closed-form/published-planning/`;
  the generator check against the closed form in `analysis/scripts/carryover-sensitivity/42-check-para`
  the earlier runs under `g10-planning/` and `g9-planning/` used other response SDs and are
  not reported;
- the paired rerun of Appendix B.4, in `analysis/scripts/carryover-sensitivity/26-co-dbc-fix-paired.`
  (run with `--params published`), with its output under `analysis/data/carryover-sensitivity/publishe`
- the check of the starting code, in `analysis/scripts/quick-sim/hendrickson-problems/14-graded-patch`

the 4,000-replicate rerun at 0.1 week in `17-published-th01-more-reps.R`, and the repair counts in `11-pd-repair-count.R`;

- the stress test, in `analysis/scripts/quick-sim/carryover-closed-form/11-strawman-stress-test.R`. The reported (third) run is reproduced with `--carryover anchored --allocation fixed --analysis-half-life 0.5`, the second with `--carryover anchored --allocation random --analysis-half-life 0.5`, and the first with `--carryover recursive --allocation fixed --analysis-half-life 0.5`, each followed by the same command with `--extra` and with `--extra2` added. The script's default analysis half-life is now 0.2 week, so the flag is needed;
- the starting code, at `github.com/rchendrickson/pmsimstats`, commit 58b32a9.

Competing interests. The authors declare no competing interests.

References

1. Sargent DJ, Conley BA, Allegra C, Collette L. Clinical trial designs for predictive marker validation in cancer treatment trials. Journal of Clinical Oncology. 2005;23(9):2020–7. doi:10.1200/JCO.2005.01.112

2. Brookes ST, Whitely E, Egger M, Davey Smith G, Mulheran PA, Peters TJ. Subgroup analyses in randomized trials: Risks of subgroup-specific analyses; power and sample size for the interaction test. Journal of Clinical Epidemiology. 2004;57(3):229–36. doi:10.1016/j.jclinepi.2003.08.009

3. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: Statistical methods and theoretical findings. Frontiers in Digital Health. 2020;2:13. doi:10.3389/fdgth.2020.00013

4. Nikles J, Mitchell GK, Schluter P, Good P, Hardy J, Rowett D, et al. Aggregating single patient (n-of-1) trials in populations where recruitment and retention was difficult: The case of palliative care. Journal of Clinical Epidemiology. 2011;64(5):471–80. doi:10.1016/j.jclinepi.2010.05.009

1407 5. Vohra S, Shamseer L, Sampson M, Bukutu C, Schmid CH, Tate R, et al. CONSORT extension for reporting N-of-1 trials (CENT) 2015 Statement. Journal of Clinical Epidemiology. 2016;76:9–17. doi:10.1016/j.jclinepi.2015.05.004

1408 6. Punja S, Xu D, Schmid CH, Hartling L, Urichuk L, Nikles CJ, et al. N-of-1 trials can be aggregated to generate group mean treatment effects: A systematic review and meta-analysis. Journal of Clinical Epidemiology. 2016;76:65–75. doi:10.1016/j.jclinepi.2016.03.026

1409 7. Senn S. Mastering variation: Variance components and personalised medicine. Statistics in Medicine. 2016;35(7):966–77. doi:10.1002/sim.6739

1410 8. Schork NJ. Personalized medicine: Time for one-person trials. Nature. 2015;520(7549):609–11. doi:10.1038/520609a

1411 9. Raskind MA, Peskind ER, Kanter ED, et al. Reduction of nightmares and other PTSD symptoms in combat veterans by prazosin: A placebo-controlled study. American Journal of Psychiatry. 2003;160(2):371–3. doi:10.1176/appi.ajp.160.2.371

1412 10. Raskind MA, Peterson K, Williams T, et al. A trial of prazosin for combat trauma PTSD with nightmares in active-duty soldiers returned from Iraq and Afghanistan. American Journal of Psychiatry. 2013;170(9):1003–10. doi:10.1176/appi.ajp.2013.12081133

1413 11. Raskind MA, Peskind ER, Chow B, et al. Trial of prazosin for post-traumatic stress disorder in military veterans. New England Journal of Medicine. 2018;378(6):507–17. doi:10.1056/NEJMoa1507598

1414 12. Raskind MA, Millard SP, Petrie EC, Peterson K, Williams T, Hoff DJ, et al. Higher pretreatment blood pressure is associated with greater posttraumatic stress disorder symptom reduction in soldiers treated with prazosin. Biological Psychiatry. 2016;80(10):736–42. doi:10.1016/j.biopsych.2016.03.2108

1415 13. Duan N, Norman D, Schmid C, Sim I, Kravitz RL. Personalized data science and personalized (N-of-1) trials: Promising paradigms for individualized health care. Harvard Data Science Review. 2022;(Special Issue 3). doi:10.1162/99608f92.8439a336

1416 14. Schmid CH, Yang J. Bayesian models for N-of-1 trials. Harvard Data Science Review. 2022;(Special Issue 3). doi:10.1162/99608f92.3f1772ce

1417 15. Wang Y, Schork NJ. Power and design issues in crossover-based N-of-1 clinical trials with fixed data collection periods. Healthcare. 2019;7(3):84. doi:10.3390/healthcare7030084

1418 16. Senn S. Cross-over trials in clinical research. 2nd ed. Chichester, UK: John Wiley & Sons; 2002.

1419 17. Morris TP, White IR, Crowther MJ. Using simulation studies to evaluate statistical methods. Statistics in Medicine. 2019;38(11):2074–102. doi:10.1002/sim.8086

1420 18. Percha B, Baskerville EB, Johnson M, Dudley JT, Zimmerman N. Designing robust N-of-1 studies for precision medicine: Simulation study and design recommendations. Journal of Medical Internet Research. 2019;21(4):e12641. doi:10.2196/12641

1421 19. Chen X, Chen P. A comparison of four methods for the analysis of N-of-1 trials. PLoS ONE. 2014;9(2):e87752. doi:10.1371/journal.pone.0087752

1422 20. Blackston JW, Chapple AG, McGree JM, McDonald S, Nikles J. Comparison of aggregated N-of-1 trials with parallel and crossover randomized controlled trials using simulation studies. Healthcare. 2019;7(4):137. doi:10.3390/healthcare7040137

1423 21. Gärtner T, Schneider J, Arnrich B, Konigorski S. Comparison of Bayesian networks, G-estimation and linear models to estimate causal treatment effects in aggregated N-of-1 trials with carry-over effects. BMC Medical Research Methodology. 2023;23(1):191. doi:10.1186/s12874-023-02012-5

1424 22. Tueller S, Ramirez D, Cance JD, Ye A, Wheeler AC, Fan Z, et al. Power analysis for idiographic (within-subject) clinical trials: Implications for treatments of rare conditions and precision medicine. Behavior Research Methods. 2023;55(8):4175–99. doi:10.3758/s13428-022-02012-1

1425 23. Grenet G, Blanc C, Bardel C, Gueyffier F, Roy P. Comparison of crossover and parallel-group designs for the identification of a binary predictive biomarker of the treatment effect. Basic and Clinical Pharmacology and Toxicology. 2020;126(1):59–64. doi:10.1111/bcpt.13293

1426 24. Van den Noortgate W, Onghena P. Hierarchical linear models for the quantitative integration of effect sizes in single-case research. Behavior Research Methods, Instruments, and Computers. 2003;35(1):1–10. doi:10.3758/BF03195492

1427 25. Moeyaert M, Ferron JM, Beretvas SN, Van den Noortgate W. From a single-level analysis to a multilevel analysis of single-case experimental designs. Journal of School Psychology. 2014;52(2):191–211. doi:10.1016/j.jsp.2013.11.003

1428 26. Moeyaert M, Yang P, Xu X. The power to explain variability in intervention effectiveness in single-case research using hierarchical linear modeling. Perspectives on Behavior Science. 2022;45(1):13–35. doi:10.1007/s40614-021-00304-z

1429 27. Senn S. Individual response to treatment: Is it a valid assumption? BMJ. 2004;329(7472):966–8.

1430 28. Araujo A, Julious S, Senn S. Understanding variation in sets of N-of-1 trials. PLoS ONE. 2016;11(12):e0167167. doi:10.1371/journal.pone.0167167

1431 29. Senn S, Rolfe K, Julious SA. Investigating variability in patient response to treatment: A case study from a replicate cross-over study. Statistical Methods in Medical Research. 2011;20(6):657–66. doi:10.1177/0962280210379174

30. Senn S. Sample size considerations for n-of-1 trials. Statistical Methods in Medical Research. 2019;28(2):372–83. doi:10.1177/0962280217726801

31. Zucker DR, Schmid CH, McIntosh MW, D'Agostino RB, Selker HP, Lau J. Combining single patient (N-of-1) trials to estimate population treatment effects and to evaluate individual patient responses to treatment. Journal of Clinical Epidemiology. 1997;50(4):401–10. doi:10.1016/S0895-4356(96)00429-5

32. Kraemer HC, Wilson GT, Fairburn CG, Agras WS. Mediators and moderators of treatment effects in randomized clinical trials. Archives of General Psychiatry. 2002;59(10):877–83.

33. Ghosh S, Henderson SG. Behavior of the NORTA method for correlated random vector generation as the dimension increases. ACM Transactions on Modeling and Computer Simulation. 2003;13(3):276–94.

34. Joe H. Generating random correlation matrices based on partial correlations. Journal of Multivariate Analysis. 2006;97(10):2177–89. doi:10.1016/j.jmva.2005.05.010

35. Jones B, Kenward MG. Design and analysis of cross-over trials. 3rd ed. Boca Raton, FL: CRC Press; 2014.

36. Senn SJ. Is the "simple carry-over" model useful? Statistics in Medicine. 1992;11(6):715–26. doi:10.1002/sim.4780110603

37. Senn S, Lambrou D. Robust and realistic approaches to carry-over. Statistics in Medicine. 1998;17(24):2849–64.

38. Holford N. Clinical pharmacology = disease progression + drug action. British Journal of Clinical Pharmacology. 2015;79(1):18–27. doi:10.1111/bcp.12170

1441 39. Galecki AT. General class of covariance structures for two or more repeated factors in longitudinal data analysis. Communications in Statistics - Theory and Methods. 1994;23(11):3105–19. doi:10.1080/03610929408831436

1442 40. Naik DN, Rao SS. Analysis of multivariate repeated measures data with a Kronecker product structured covariance matrix. Journal of Applied Statistics. 2001;28(1):91–105. doi:10.1080/02664760120011626

1443 41. Li F, Chen X, Tian Z, Esserman D, Heagerty PJ, Wang R. Designing three-level cluster randomized trials to assess treatment effect heterogeneity. Biostatistics. 2023;24(4):833–49. doi:10.1093/biostatistics/kxac026

1444 42. Hobbs DC, Twomey TM, Palmer RF. Pharmacokinetics of prazosin in man. Journal of Clinical Pharmacology. 1978;18(8-9):402–6. doi:10.1002/j.1552-4604.1978.tb02456.x

1445 43. Holford NHG, Sheiner LB. Understanding the dose-effect relationship: Clinical application of pharmacokinetic-pharmacodynamic models. Clinical Pharmacokinetics. 1981;6(6):429–53. doi:10.2165/00003088-198106060-00002

1446 44. Kuznetsova A, Brockhoff PB, Christensen RHB. lmerTest package: Tests in linear mixed effects models. Journal of Statistical Software. 2017;82(13):1–26. doi:10.18637/jss.v082.i13

1447 45. Siepe BS, Bartoš F, Morris TP, Boulesteix AL, Heck DW, Pawel S. Simulation studies for methodological research in psychology: A standardized template for planning, preregistration, and reporting. Psychological Methods. 2024. doi:10.1037/met0000695

1448 46. pmsimstats team. Carryover representation and inference for biomarker-treatment interaction tests with a predictive biomarker in aggregated N-of-1 trials under covariance moderation: A staged simulation study and a recommended analysis for the Hybrid design. 2026.

1449 47. Pawel S, Kook L, Reeve K. Pitfalls and potentials in simulation studies: Questionable research practices in comparative simulation studies allow for spurious claims of superiority of any method. Biometrical Journal. 2024;66(1):2200091. doi:10.1002/bimj.202200091

1450 48. Schork NJ. Accommodating Serial Correlation and Sequential Design Elements in Personalized Studies and Aggregated Personalized Studies. Harvard Data Science Review. 2022;(Special Issue 3). doi:10.1162/99608f92.f1eef6f4

1451 49. Haller B, Ulm K. A simulation study on estimating biomarker-treatment interaction effects in randomized trials with prognostic variables. Trials. 2018;19(1):128.

1452 50. Kent DM, Paulus JK, Klaveren D van, et al. The predictive approaches to treatment effect heterogeneity (PATH) statement. Annals of Internal Medicine. 2020;172(1):35–45. doi:10.7326/M18-3667

1453 51. Kenward MG, Roger JH. Small sample inference for fixed effects from restricted maximum likelihood. Biometrics. 1997;53(3):983–97. doi:10.2307/2533558

1454 52. Bell RM, McCaffrey DF. Bias reduction in standard errors for linear regression with multi-stage samples. Survey Methodology. 2002;28(2):169–81.

1455 53. Pustejovsky JE, Tipton E. Small-sample methods for cluster-robust variance estimation and hypothesis testing in fixed effects models. Journal of Business and Economic Statistics. 2018;36(4):672–83. doi:10.1080/07350015.2016.1247004

1456 54. Ding P, Feller A, Miratrix L. Decomposing treatment effect variation. Journal of the American Statistical Association. 2019;114(525):304–17. doi:10.1080/01621459.2017.1407322

1457 55. Bauer DJ, Curran PJ. Distributional assumptions of growth mixture models: Implications for overextraction of latent trajectory classes. Psychological Methods. 2003;8(3):338–63. doi:10.1037/1082-989X.8.3.338

1458 56. Lubke GH, Muthén B. Investigating population heterogeneity with factor mixture models. Psychological Methods. 2005;10(1):21–39. doi:10.1037/1082-989X.10.1.21

1459 57. Nylund KL, Asparouhov T, Muthén BO. Deciding on the number of classes in latent class analysis and growth mixture modeling: A Monte Carlo simulation study. Structural Equation Modeling. 2007;14(4):535–69. doi:10.1080/10705510701575396

1460 58. Liu GF, Lu K, Mogg R, Mallick M, Mehrotra DV. Should baseline be a covariate or dependent variable in analyses of change from baseline in clinical trials? Statistics in Medicine. 2009;28(20):2509–30. doi:10.1002/sim.3639

1461 59. Higham NJ. Computing a nearest symmetric positive semidefinite matrix. Linear Algebra and its Applications. 1988;103:103–18. doi:10.1016/0024-3795(88)90223-6

1462 60. Higham NJ, Strabić N, Šego V. Restoring definiteness via shrinking, with an application to correlation matrices with a fixed block. SIAM Review. 2016;58(2):245–63. doi:10.1137/140996112

Appendix A. Closed-form derivations

A.1 Moments of the paired-difference statistic

Let $a_t = 1/n_{\text{on}}$ at on-drug visits and $-1/n_{\text{off}}$ at off-drug visits, so that $\Delta_i = \sum_t a_t Y_{it}$. Because the weights sum to zero, the baseline level cancels and $\Delta = -\sum_t a_t S_t$, with $S_t = TV_t + PB_t + BR_t$. In general $\text{Var}(\Delta) = a^\top \Sigma_S a$, where Σ_S is the covariance of the summed response, whatever the structure.

With $m_1 = \sum a_t$, $m_2 = \sum a_t^2$, $m_\eta = \sum a_t \eta_t$, $m_{\eta\eta} = \sum a_t^2 \eta_t^2$ and $m_{2\eta} = \sum a_t^2 \eta_t$, with η_t the design expectancy, the variance under compound symmetry with common correlation ρ_{cs} is

$$\begin{aligned}\text{Var}(\Delta) = {} & (\sigma_{TV}^2 + \sigma_{BR}^2)[\rho_{cs} m_1^2 + (1-\rho_{cs}) m_2] + \sigma_{PB}^2 [\rho_{cs} m_\eta^2 + (1-\rho_{cs}) m_{\eta\eta}] \\ & + 2\sigma_{TV}\sigma_{BR}[c_\times m_1^2 + (c_1 - c_\times) m_2] + 2(\sigma_{TV} + \sigma_{BR})\sigma_{PB}[c_\times m_1 m_\eta + (c_1 - c_\times) m_{2\eta}],\end{aligned}$$

where c_1 and $c_\times$ are the same-visit and other-visit cross-component correlations. Under the separable form of Stage 5, $\text{Cov}(S_t, S_s) = (s_t^\top K s_s) A_{ts}$ (A.5), and $\text{Var}(\Delta) = \sum_{t,s} a_t a_s (s_t^\top K s_s) A_{ts}$, with s_t the vector of component standard deviations at visit t.

- **Proposition 1.** No term depends on $t_{1/2}$ under either structure, so $\text{Var}(\Delta)$ is free of the half-life. The argument applies to any correlation matrix in which the half-life does not enter the response block. It therefore holds for unrepaired matrices only. The repair changes the response block, and its changes depend on the half-life through the correlation profile. For E9 the weights sum to zero, so $m_1 = 0$. For E9b the post-baseline weights sum to $1/(n_{\text{off}} + 1)$, and the baseline row, with $S_0 = 0$, carries the remaining weight.
- **Proposition 2.** Only BR_t is correlated with B, with covariance $c_{bm} g_t \sigma_{bm} \sigma_{BR}$, so $\beta_{bm:D} = -c_{bm}(\sigma_{BR}/\sigma_{bm}) \sum_t a_t g_t$. With $g_t = 1$ on drug, this gives the expressions in the Methods.
- **Residual variance.** Under the correlation encoding, $\sigma_\Delta^2 = \text{Var}(\Delta) - \beta_{bm:D}^2 \sigma_{bm}^2$.

A.2 Power

Within a path of n participants, the least-squares slope is exactly normal given the biomarker values. Its t-statistic is exactly noncentral t with $n-2$ degrees of freedom and noncentrality $\beta_{bm:D}\sqrt{\text{SS}_B}/\sigma_\Delta$,

where $\mathrm{SS}_B \sim \sigma^2_{bm}\chi^2_{n-1}$.

At $\mathrm{SS}_B = \sigma^2_{bm}(n-1)$ the noncentrality is $\Lambda = \sqrt{n-1}\, r_{\Delta B}/\sqrt{1-r^2_{\Delta B}}$, where $r_{\Delta B} = \beta_{bm:D}\sigma_{bm}/\mathrm{SD}(\Delta)$ is proportional to the on/off correlation difference. For small $r_{\Delta B}$ the noncentrality is approximately $\sqrt{n-1}\, r_{\Delta B}$. The sample size for a fixed noncentrality therefore scales as $1/r^2_{\Delta B}$ and hence approximately as $1/\mathcal{G}^2$. The exact scaling uses $r/\sqrt{1-r^2}$.

Pooling paths with a common intercept adds the between-path variation in mean contrast to the residual variance. In the Hybrid design it is under 0.7% of the total. Power is the noncentral-t tail probability averaged over the χ^2 law of SS_B. For the pooled statistic this is an approximation. It treats the between-path differences in mean and slope as extra residual variance and ignores the differences in residual variance between paths. The per-path result is exact.

A.3 Monotonicity under the decaying correlation

$$\frac{d\bar{g}_{\text{off}}}{dt_{1/2}} = \frac{\ln 2}{n_{\text{off}}\, t^2_{1/2}} \sum_{t\,\text{off}} t_{sd,t}\, 2^{-t_{sd,t}/t_{1/2}} > 0$$

on any path with at least one off-drug visit after an exposure.

- **Proposition 3.** On such a path, by Proposition 2 the slope's magnitude is strictly decreasing in $t_{1/2}$. By A.1, σ^2_Δ is strictly increasing, since $\mathrm{Var}(\Delta)$ is fixed. So the noncentrality and the power of the path-stratified statistic are strictly decreasing. On a path whose off-drug visits all precede first exposure (the placebo-first crossover path), $\bar{g}_{\text{off}} = 0$ at every half-life and the power is constant. The pooled statistic's residual variance also contains the between-path variation of mean contrasts, which depends on the half-life through the carryover means. Its monotonicity was checked on a grid (Stage 2), not proved.
- **Small half-lives.** $\bar{g}_{\text{off}} \le 2^{-1/t_{1/2}}$, which gives the behavior near zero stated in the Results.
- **Large half-lives.** $\bar{g}_{\text{off}} \to 1$ as $t_{1/2} \to \infty$, which recovers the constant rule.

A.4 The feasibility bound and the shared biomarker effect

The feasibility bound. Order the variables as biomarker, then responses, and partition the correlation matrix as $\begin{pmatrix} 1 & c_{bm}\tilde{g}^\top \\ c_{bm}\tilde{g} & M \end{pmatrix}$. By the Schur complement, the matrix is positive definite if and only if M is positive definite and $1 - c^2_{bm}\tilde{g}^\top M^{-1}\tilde{g} > 0$. Hence the bound is $c^*_{bm} = (\tilde{g}^\top M^{-1}\tilde{g})^{-1/2}$.

With prognostic correlations. Let M_0 be the correlation matrix of everything but the biomarker, including baseline severity, which is uncorrelated with the response components. The biomarker row is a sum of fixed patterns scaled by the correlations, $r = c_{bm}\tilde{g} + c_{PB}\,p + c_{TV}\,q + c_{BL}\,b$, where p and q are one on every visit of the placebo-response and natural-history components and b is one on baseline severity. The same Schur argument gives the condition $r^\top M_0^{-1} r < 1$, that is $c^\top A c < 1$ with $c = (c_{bm}, c_{PB}, c_{TV}, c_{BL})$ and $A_{jk} = x_j^\top M_0^{-1} x_k$ for $x \in \{\tilde{g}, p, q, b\}$. The feasible region is the interior of an ellipsoid. The bound for a single correlation is the case $A_{\tilde{g}\tilde{g}}^{-1/2}$. Fixing all correlations but one makes $c^\top A c = 1$ a quadratic in that correlation, whose positive root is its bound. A negative discriminant means the fixed correlations are already infeasible. Because b is an eigenvector of M_0 with unit eigenvalue, c_{BL} enters only through its square. Its bound given c_{bm} is $\sqrt{1 - c_{bm}^2 A_{\tilde{g}\tilde{g}}}$. The p and q patterns are structurally identical, so c_{PB} and c_{TV} have the same bounds. A design's bound is the minimum over its paths. The off-diagonal entries of A are negative at the recommended DGP's values, so the correlations are mildly complementary. The DGP sets the biomarker's marginal correlations with the other components to zero. The negative entries of A reflect partial association: given its correlation with the drug response, the biomarker's partial correlation with the other components is not zero. An explicit correlation of the same sign therefore uses less of the feasible region than it would if the components were independent.

The shared biomarker effect. Let s_0, s_{off} and s_{on} be the true slopes of Y on B at baseline, at post-baseline off-drug visits and at on-drug visits. A model with baseline as an off-drug row estimates one β_B for s_0 and s_{off}.

The following account is heuristic, not a derivation. Consider a prognostic biomarker with no interaction, whose post-baseline slope is constant: $s_0 = 0 \neq s_{\text{off}} = s_{\text{on}} = s_{\text{prog}}$.

- **The bias.** The fitted $\hat{\beta}_B$ lies between the baseline and off-drug slopes, $\hat{\beta}_B \approx f\, s_{\text{prog}}$ for some $0 < f < 1$ that depends on the weight the baseline row receives. The interaction absorbs the remainder, $\hat{\beta}_{bm:D} \approx (1 - f)\, s_{\text{prog}} \neq 0$.
- **The correction.** Adding $B_i \cdot \mathbf{1}\{t = 0\}$ frees s_0 and removes this source of bias. The constraint is analogous to the choice, in longitudinal analysis, between treating baseline as a covariate and as a response (58).

Two further mechanisms operate in the stress-test scenarios.

- **The slope grows over time.** When the post-baseline slope grows, the bias depends on how

drug status is confounded with time. Freeing s_0 is then not sufficient in general. The rejection rate of 0.066 for that analysis under the growing pathway is consistent with this.

- **Expectancy.** When the biomarker is associated with the placebo response, the slope also scales with the expectancy η_t, which is higher at the open-label (on-drug) visits. The expectancy term $B_i \cdot \eta_t$, with $\eta_0 = 0$, addresses both this mechanism and the baseline constraint. Freeing s_0 alone does not. An expectancy-scaled slope still enters the interaction through the open-label visits.

Neither correction is exact in the Hybrid design. The expectancy term removes a constant post-baseline slope from the interaction only when the blinded on-drug and off-drug visits carry equal total weight. In the Hybrid design they do not, so a small bias remains. The corresponding derivations are in the project's compendium of mathematical results (Additional file 5). The stress-test null rejection rates across the four prognostic cells were 0.040 to 0.055 for the expectancy-term model and 0.051 to 0.066 for the free-baseline model. The residual bias of the expectancy-term model is therefore within Monte Carlo error at the prognostic correlations simulated. That of the free-baseline model is not, under the growing pathway.

A scenario in which the biomarker is associated with baseline severity alone was simulated in the stress test. The binary-indicator analysis rejected 0.064 there. The two corrections rejected 0.055 and 0.054.

A.5 The separable structure

Let $A_{ts} = \rho^{|w_t - w_s|}$ and $K = (1 - c_1)I_3 + c_1 J_3$, and set $M = K \otimes A$, so that the correlation between component c at visit t and component c' at visit s is $K_{cc'}A_{ts}$.

Validity. The eigenvalues of M are the products of those of K and A.

- K has eigenvalues $1 + 2c_1$ and $1 - c_1$ (twice), so it is positive definite for $-\frac{1}{2} < c_1 < 1$.
- A is the correlation matrix of a stationary Ornstein-Uhlenbeck process at the visit times, and is positive definite for any distinct times.

Hence M is positive definite for every schedule. Generatively, $\mathbf{X} = L_A \mathbf{Z} L_K^\top$ with $L_A L_A^\top = A$ and $L_K L_K^\top = K$: each column of $L_A \mathbf{Z}$ is an independent AR(1) path, and the components are fixed mixtures of them.

The feasibility bound. With the biomarker correlated with the drug-response component only, its correlation vector is $c_{bm}\tilde{g}$, with $\tilde{g} = e_3 \otimes g$ and g the correlation profile. Since $M^{-1} = K^{-1} \otimes A^{-1}$,

$$\tilde{g}^\top M^{-1} \tilde{g} = [K^{-1}]_{33}\, g^\top A^{-1} g, \qquad [K^{-1}]_{33} = \frac{1 + c_1}{(1 - c_1)(1 + 2c_1)} = \frac{1}{1 - R^2}.$$

The schedule term. The sampled process is Markov, with step correlation $\rho_t = \rho^{w_t - w_{t-1}}$, so

$$g^\top A^{-1} g = g_1^2 + \sum_{t=2}^{n} \frac{(g_t - \rho_t g_{t-1})^2}{1 - \rho_t^2}.$$

For a binary correlation profile, each step from visit $t-1$ to visit t contributes the following term:

Step	Term
on to on	$(1 - \rho_t)/(1 + \rho_t)$
on to off	$\rho_t^2/(1 - \rho_t^2)$
off to on	$1/(1 - \rho_t^2)$
off to off	0

Under the decaying correlation, the off-drug correlation decays instead of switching, which lowers the on-to-off term. This is why the feasibility bound under the decaying correlation rises with the half-life. Under compound symmetry it rises from 0.256 without carryover to 0.307 at 1 week and 0.369 at 2 weeks, which makes $c_{bm} = 0.3$ feasible from 1 week (Figure 5).

The two-constant form. With a common ρ, setting the same-visit cross-component correlation to c_1 and the different-visit correlation to $c_\times A_{ts}$ gives

$$M = K_a \otimes A + K_b \otimes I_n, \qquad K_a = (1 - c_\times) I_3 + c_\times J_3, \qquad K_b = (c_1 - c_\times)(J_3 - I_3).$$

K_b has zero diagonal and so is indefinite. The eigenvalues of M are $\omega_j(A)(1 + 2c_\times) + 2(c_1 - c_\times)$ and $\omega_j(A)(1 - c_\times) - (c_1 - c_\times)$, where $\omega_j(A)$ are the eigenvalues of A. Hence M is positive definite if and only if $\omega_{\min}(A) > (c_1 - c_\times)/(1 - c_\times)$. For visits at a common spacing of s weeks, $\omega_{\min}(A)$ decreases toward $(1 - \rho^s)/(1 + \rho^s)$ as visits accumulate, which is 0.111 at weekly spacing and $\rho = 0.8$ (the threshold itself) and 0.016 at daily spacing. The values for the paper's schedules are in Stage 5.

The starting structure. The starting compound-symmetry form is $M = K_o \otimes I_n + K_p \otimes J_n$, with a person-level factor $K_p = \rho_{cs} I_3 + c_\times (J_3 - I_3)$ and an occasion-level factor $K_o = (1 - \rho_{cs}) I_3 + (c_1 - c_\times)(J_3 - I_3)$. It is positive definite if and only if $(1 - \rho_{cs}) - (c_1 - c_\times) > 0$, whatever the schedule (0.10 at the starting values, which is also the smallest eigenvalue of the eight-visit response block of the starting DGP).

The summed response. With component standard deviations s_t and $S_t = \sum_c s_{c,t} X_{c,t}$, separability gives

$$\mathrm{Cov}(S_t, S_s) = (s_t^\top K s_s)\, A_{ts},$$

proportional to the AR(1) kernel when the standard deviations are constant. The correlation of the summed response is

$$\mathrm{Corr}(S_t, S_s) = q_{ts}\, A_{ts}, \qquad q_{ts} = \frac{s_t^\top K s_s}{\sqrt{(s_t^\top K s_t)(s_s^\top K s_s)}} \le 1,$$

by the Cauchy-Schwarz inequality in the inner product defined by the positive definite K. Equality holds if and only if s_s is a multiple of s_t. In the Hybrid design $s_t = (\sigma_{TV}, \sigma_{PB}\eta_t, \sigma_{BR}) = (10, 10\eta_t, 8)$. Within a phase η_t is constant, so $q_{ts} = 1$ and the summed response is exactly AR(1) there. Between an open-label visit ($\eta_t = 1$) and a blinded visit ($\eta_s = 0.5$), with $c_1 = 0.2$, $s_t^\top K s_t = 368$, $s_s^\top K s_s = 257$, $s_t^\top K s_s = 300$ and $q_{ts} = 0.976$. The summed response therefore has a variance that changes at blinding and a cross-phase correlation 2.4% below the AR(1) value. A constant-variance `corCAR1` model omits both. The analyses with phase-specific residual variances in the stress test represent the first and not the second. The two-constant form adds $(s_t^\top K_b s_t)\, \mathbf{1}[t = s] > 0$, a white-noise term.

These results are derived in full, and checked numerically, in the pmsimstats-ng documentation (Additional file 6).

A.6 The responder-class reading of c_{bm}

Let $Z_i \in \{0, 1\}$ be a latent responder class with $\Pr(Z_i = 1) = p$, and let $B_i = \mu_B(Z_i) + e_{B,i}$ and $BR_i = \mu_{BR}(Z_i) + e_{BR,i}$ for the on-drug drug response, with e_B and e_{BR} independent of each other and of Z. Then $\mathrm{Cov}(B, BR) = \mathrm{Cov}(\mu_B(Z), \mu_{BR}(Z))$. Both $\mu_B(Z)$ and $\mu_{BR}(Z)$ are affine in the binary Z, so their correlation is ± 1 and $|\mathrm{Cov}(B, BR)| = \mathrm{SD}(\mu_B(Z))\, \mathrm{SD}(\mu_{BR}(Z))$. Dividing by

$\sigma_B \sigma_{BR}$,

$$c_{bm}^2 = \frac{\mathrm{Var}(\mu_B(Z))}{\sigma_B^2} \cdot \frac{\mathrm{Var}(\mu_{BR}(Z))}{\sigma_{BR}^2} = f_B \, f_{BR},$$

the product of the between-class variance fractions. With more than two classes the correlation of $\mu_B(Z)$ and $\mu_{BR}(Z)$ is at most 1 in magnitude, so $c_{bm}^2 \le f_B f_{BR}$.

Appendix B. Code changes

B.1 The decaying correlation in the starting `generateData()` (commit 58b32a9)

The starting code (`R/generateData.R`, lines 131 to 138) set the biomarker-response correlation to c_{bm} wherever the drug-response mean is nonzero:

```
# correlation with biomarker
for(p in 1:nP){
  n1<-paste(trialdesign$timeptname[p],"br",sep=".")
  if(means[which(n1==labels)]!=0){
    correlations[n1,'bm']<-modelparam$c.bm
    correlations['bm',n1]<-modelparam$c.bm
  }
}
```

The replacement sets the correlation in proportion to the remaining drug effect, with zero correlation before first exposure:

```
# correlation with biomarker: full on drug, decaying with the drug
# effect off drug (decaying correlation), zero before first exposure
for(p in 1:nP){
  n1<-paste(trialdesign$timeptname[p],"br",sep=".")
  if(d[p]$tod>0){
    g<-1
  }else if(d[p]$tsd>0 && modelparam$carryover_t1half>0){
```

```
      g<-(1/2)^(d[p]$tsd/modelparam$carryover_t1half)
    }else{
      g<-0
    }
    correlations[n1,'bm']<-modelparam$c.bm*g
    correlations['bm',n1]<-modelparam$c.bm*g
  }
```

The constant rule tests whether the off-drug drug-response mean is nonzero. In double precision that mean underflows to zero at late off-drug visits when the half-life is below about 0.007 week, or below about 0.012 week under the compounding carryover rule. The constant rule then sets $g_t = 0$ at those visits. This is a property of the floating-point arithmetic, not of the DGP. It is why Stage 2 describes power under the constant rule as constant at every positive half-life of practical interest.

B.2 Stop instead of repairing

The starting code replaces a non-positive-definite covariance matrix with a nearby valid one when its `makePositiveDefinite` option is on, as it is in the starting driver (lines 145 to 150). It calls `corpcor::make.positive.definite` with tolerance 10^{-3}. That function follows Higham's nearest symmetric positive semidefinite matrix (59). It raises every eigenvalue below 2×10^{-3} to that value, so it changes the variances as well as the correlations. No warning is issued. The replacement below stops instead, so that an infeasible cell is reported rather than changed:

```
  # published:
  # if(makePositiveDefinite){
  #   if(!is.positive.definite(sigma)){
  #     sigma<-make.positive.definite(sigma, tol=1e-3)
  #   }
  # }
  if(!corpcor::is.positive.definite(sigma)){
    stop(sprintf('c.bm = %.3f exceeds the feasible range for this design',
                 modelparam$c.bm))
  }
```

One case is exempt from the stop. The placebo-response component's standard deviation is proportional to the expectancy. A visit with expectancy zero (a run-in or a baseline visit coded with no expectancy) has exactly zero placebo-response variance. It makes Σ singular by construction rather than by an infeasible correlation. The package's stop path checks and factors only the sub-block of positive-variance components and draws the zero-variance components at their means. No correlation is modified. The repair set those variances to 10^{-3} instead.

How often the starting grid needed repair. The nonzero effect sizes of the earlier grid were 0.3 and 0.6. Counting one matrix per design path, effect size and half-life, 20 of its 54 matrices at those effect sizes (37%) were replaced by nearby valid matrices (Table 16). At $c_{bm} = 0.3$ the repair changed no correlation by more than 0.007. Results at that effect size are therefore essentially those of the stated DGP. At $c_{bm} = 0.6$ the repair lowered the on-drug correlation to 0.54 to 0.57 and moved other correlations by up to 0.10. Those results describe a different DGP. The constant rule passes more cells at positive half-lives, because a constant correlation places a small demand on the structure. It also carries no interaction (Stage 2). The cells that need no repair under the constant rule are therefore the cells in which the interaction is absent.

A repair need not be the unstructured nearest matrix. A shrinking repair can keep the response block fixed and change only the biomarker row (60). Such a repair simulates a smaller c_{bm} than the one requested. The same feasibility question arises for target correlations in NORTA generation (33). Correlation matrices built from partial correlations are positive definite by construction (34), but they do not fix the target correlations a design requires. The stop is the simpler choice. It reports the infeasibility to the user.

The case that fixed the default was a prognostic scenario in the companion study (46). Two runs combined $c_{bm} = 0.45$ with a prognostic correlation of 0.6. The joint bound of Stage 6 shows that combination to be infeasible at any c_{bm}. The repair lowered the on-drug correlation, and the runs reported the attenuated interaction as a loss of power to the prognostic pathway. A repair can therefore turn an infeasible input into a plausible but wrong finding.

B.3 Carryover decaying from the last on-drug mean

The compounding carryover rule (lines 83 to 92) decays the previous visit's mean by the total time since discontinuation. At the second and later off-drug visits that mean has already been decayed, so the decay compounds. The replacement decays from the last on-drug value. Study 1 uses it at

Table 16: Correlation matrices that are not positive definite. Figure 4 grid: open label, open label plus blinded discontinuation, crossover and Hybrid designs; $t_{1/2} \in \{0, 0.1, 0.2\}$ week. Extended grid: the last three designs; $t_{1/2} \in \{0, 0.1, 0.2, 0.5, 1\}$ week. The starting-DGP counts reproduce those of the starting code exactly. The recommended-DGP counts are the same at $\rho = 0.7$ (the recommended value) and $\rho = 0.8$ because both feasibility bounds lie between 0.3 and 0.6.

Grid (number of matrices)	Starting DGP (constant correlation, compound symmetry, $\rho_{cs} = 0.8$)	Recommended DGP (decaying correlation, separable AR(1)), $\rho = 0.8$	Recommended DGP, $\rho = 0.7$
Figure 4 grid of the earlier study, $c_{bm} \in \{0, 0.3, 0.6\}$ (81)	20 (25%)	21 (26%)	21 (26%)
Extended grid, $c_{bm} = 0.25$ (40)	0 (0%)	0 (0%)	0 (0%)
Extended grid, $c_{bm} = 0.3$ (40)	12 (30%)	0 (0%)	0 (0%)
Extended grid, $c_{bm} = 0.6$ (40)	12 (30%)	30 (75%)	30 (75%)

every positive half-life. Replacing it with the constant rule changes no slope, variance or feasibility bound, only the path means. It therefore changes the study 1 closed-form power by at most 0.002 in the Hybrid design. In the starting pipeline the change alone moves power by at most 0.006 (Additional file 7, Section 8.3). Combined with the restart of the Gompertz clock at re-exposure, it can produce a downward step in the mean at the first re-exposure visit when the residual exceeds the restarted curve's value. On the Hybrid schedule it does not (Stage 3).

```
brmeans<-modgompertz(d$tod,rp$max,rp$disp,rp$rate)
if(nP>1){
  last_on<-0
  for(p in 1:nP){
    if(d[p]$onDrug){
      last_on<-brmeans[p]
    }else if(d[p]$tsd>0 && modelparam$carryover_t1half>0){
      brmeans[p]<-last_on*(1/2)^(d$tsd[p]/modelparam$carryover_t1half)
    }
  }
```

```
}
```

B.4 The exposure-decayed drug indicator in the analysis

Analyses that use an exposure-decayed drug indicator must code off-drug visits before first exposure as unexposed. The time since discontinuation is zero there, and the decay function returns 1 at zero, so without a guard a placebo-first crossover participant would be coded fully exposed at every visit. In the pmsimstats-ng carryover manuscript this guard was missing. In a paired rerun of that manuscript's crossover cells (`analysis/scripts/carryover-sensitivity/26-co-dbc-fix-paired.R`, starting parameters, $N = 70$, $c_{bm} = 0.45$, $t_{1/2} = 1$ week, 500 replicates, model-based test, analysis half-life equal to the true one), restoring the guard raised the power of the exposure-weighted analysis from 0.302 to 0.534, within Monte Carlo error of the binary indicator (0.536):

```
exposure_weight <- function(Db, tsd, halflife) {
  w <- 0.5^(tsd / halflife)
  ifelse(Db == 1, 1, ifelse(tsd > 0, w, 0))
}
```

B.5 The recommended DGP in pmsimstats-ng

The package function `buildSigma()` (`R/generateData.R`) implements the decaying correlation, with `c.bm` on drug, `c.bm * exp(-lambda_cor * tsd)` off drug after exposure and zero before exposure. It implements AR(1) within each component in calendar weeks. It decays carryover on the drug-response mean from the last on-drug value (not the constant rule of B.3). It implements the prognostic correlations of Stage 6: with the placebo-response component (`c.bm.pb`), with the natural-history component as a constant correlation (`c.bm.tv`) or as a mean shift growing along the natural-history curve (`b.bm.tv`), and with baseline severity (`c.bm.bl`). The first three enter the biomarker row of the correlation matrix and share the joint bound of Table 9. The mean shift does not enter the matrix and has no bound.

The cross-component block. `buildSigma()` builds the cross-component correlation from two constants: `c.cf1t` at the same visit and `c.cfct * rho^tg` at a lag of `tg` weeks. With `c.cf1t = c.cfct = c_1` this is exactly the separable form of Table 11, $c_1 \rho^{|w_t - w_s|}$ at every lag, since $\rho^0 = 1$.

With the two constants unequal it is the two-constant form of Stage 5. The recommended DGP is therefore the package with both constants set to 0.2, which is how the project's carryover driver calls it (`analysis/scripts/carryover-sensitivity/simulation-core.R`). Study 3's driver builds the separable block directly (`C1 * A` in `11-strawman-stress-test.R`).

Allocation and carryover in study 3. The driver takes `--allocation fixed|random` and `--carryover anchored|recursive`. The first run used fixed allocation (18, 18, 17, 17) and the compounding carryover rule (`recursive`). The second run used carryover decaying from the last on-drug mean (`anchored`) and randomized allocation. The third run used the decaying rule and fixed allocation, the recommended DGP as defined. All three used the same seed rule and an analysis half-life of 0.5 week. The third run changes one setting relative to each earlier run, so the effects of the carryover rule and of the allocation can be read separately (Stage 6). The tables in Stage 6 are from the third run, and the first run's prognostic table is in Appendix C. Randomized allocation is a driver option that is not part of the definition.

Stop instead of repair. `buildSigma()`, `generateData()` and `generateSimulatedResults()` accept a `stopOnNonPD` argument, `TRUE` by default. A non-positive-definite covariance matrix then raises an error, exactly as in B.2, both at the explicit positive-definiteness check and in the Cholesky-factorization fallback that previously repaired regardless of the `makePositiveDefinite` setting. The starting DGP's repair is reproduced only by passing `stopOnNonPD = FALSE` and setting `options(pmsimstats.allow_repair = TRUE)` explicitly, as the scripts that study the repair itself do. Without the option, the repair is locked off: a draw from a matrix that is not positive definite stops with an error whatever the other arguments are. The argument was first added with a `FALSE` default for backward compatibility. The default was changed after two runs of the companion carryover study repaired infeasible cells and reported the attenuated interaction as a power loss.

Not in the package. The measurement-error, Weibull-decay and dropout options of Table 11 are implemented in the stress-test driver (`11-strawman-stress-test.R`) and in the companion study's driver core, and not yet in the package. The package's dropout mechanisms (`censordata.R`) are the starting censoring patterns, with one correction. In the starting code the biased term weights each participant by 100 plus the improvement from baseline, raised to the power `eb1`. Dropout therefore rose with improvement, the reverse of the mechanism the earlier study describes. The package now weights by 100 minus the improvement, floored at zero, so that dropout rises with worsening. Figure 4 uses the corrected rule. Figure 3 and `implementations/original` keep the

Table 17: Null rejection rates under a prognostic biomarker in the first run of study 3 (compounding carryover rule, fixed allocation), 1,000 replicates per cell (two-MCSE band 0.036 to 0.064 for a single cell). Rates above 0.075 in bold.

Analysis	Natural history, constant	Natural history, growing	Placebo response	Baseline severity
Binary-indicator analysis (random intercept, model-based)	**0.107**	0.053	**0.159**	0.075
Random intercept + CAR(1), CR2	**0.099**	0.049	**0.100**	0.050
The same, half-life chosen by AIC	**0.115**	0.058	**0.112**	0.055
Unstructured covariance, Kenward-Roger	**0.416**	**0.195**	**0.222**	0.051
The same, half-life chosen by AIC	**0.509**	**0.245**	**0.256**	0.064
Random intercept + CAR(1) with expectancy terms, CR2	0.047	0.046	0.040	0.040
The same, with phase-specific residual variances, CR2	0.055	0.044	0.042	0.042
The same, with phase-specific residual variances, model-based	0.062	0.043	0.045	0.046
Expectancy terms, half-life chosen by AIC, CR2	0.059	0.052	0.047	0.045
Random intercept + CAR(1) with free baseline slope, CR2	0.049	0.063	0.049	0.050

published rule.

Appendix C. First run of the stress test

The first run of study 3 used the compounding carryover rule on the mean and fixed path allocation (18, 18, 17, 17). Its prognostic table is given here for comparison with Table 8. The two runs differ only in the carryover rule (Methods).